# The Effects of US Clean Electricity Subsidies on Infrastructure, Emissions, and Welfare

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**Abstract**

Utility-scale clean electricity subsidies have long been a centerpiece of US federal climate policy. I estimate the effects of clean electricity subsidies on electricity capacity, emissions, and welfare by exploiting the spatial variation in subsidy levels created by a recent place-based subsidy, the Energy Community Bonus. The additional subsidy causes large increases in solar and battery capacity, with smaller gains in wind capacity. This translates to significant emissions abatement and welfare gains. The Energy Community Bonus was enacted to achieve a distributional objective, but it incidentally also changes how subsidies affect emissions and welfare. Whether the place-based subsidy presents an efficiency-equity tradeoff or win-win depends on the subsidized technology. The Energy Community Bonus improves the welfare and emissions effects of solar and wind subsidies, creating a combined $800 million per year of surplus. However, it reduces the welfare and emissions effects of battery subsidies by $1.1 billion per year.

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For decades, utility-scale clean electricity subsidies have been the most fiscally expensive US policy to address climate change. From 2000 to 2022, the subsidies had a fiscal cost of over $130 billion.[1] Recent years have seen large, rapid shifts in clean electricity subsidy policy. In 2022, the Inflation Reduction Act (IRA) expanded the subsidies to an estimated total fiscal cost of $400 billion through 2035 (Bistline et al., 2023a). Then, in 2025, the One Big Beautiful Bill Act (OBBBA) eliminated the subsidies for solar and wind, which account for about half of future planned clean capacity. It kept the subsidies in force for the other half, mostly batteries. Despite the enormity of these policies, little is known about their physical and welfare effects. This is because there has historically been no quasi-experimental variation in subsidy levels that could be used to identify their effects. Pundits speculate that these subsidies do little to hasten the deployment of clean electricity and are largely or entirely inframarginal, citing permitting delays, local opposition to nearby generators, and institutional barriers (NYT, 2023; WSJ, 2024). Renewable electricity companies and interest groups claim they do not need the subsidies to be profitable, while policymakers insist the subsidies will be re-enacted anyway (Politico, 2026). Meanwhile, models of the electricity sector like the one that produced the $400 billion estimate suggest clean electricity subsidies have large effects and are the single most impactful US federal climate policy (Bistline et al., 2023b).

Part of the IRA reform, the "Energy Community Bonus," made clean electricity subsidies place-based. The Energy Community Bonus offers a higher-value subsidy to generators located in "Energy Communities," areas with local economies reliant on fossil fuels. I use this place-based subsidy variation to estimate the effect of a marginal subsidy increase on electricity capacity. I combine the empirical estimates with electricity and emissions data in a sufficient statistics framework to model emissions and welfare effects of clean electricity subsidies.

The same place-based provision that provides the variation for this exercise also changes the answer to the question of welfare effects. The usual economic intuition for combining distributional mechanisms with externality pricing is that there is an efficiency-equity trade-off. However, that intuition only applies to direct externality prices. Clean electricity subsidies are per megawatt (MW) or megawatt-hour (MWh), not per unit of externality, and the externality generated by a MW or MWh depends on where the electricity is generated. So, the place-based subsidy could improve or worsen subsidy welfare effects, depending on whether clean electricity happens to abate more or less emissions in Energy Communities than it does elsewhere. I conclude with an assessment of whether the place-based provision poses an efficiency-equity tradeoff.

[1] In 2025 USD, calculated from Joint Committee on Taxation and Section 1603 Grant reports.

The seemingly straightforward question of how a subsidy affects new generator construction is complicated by the short lifespan of the Energy Community Bonus for solar and wind and the long timeline required to build a new generator in the US. The Energy Community Bonus took effect in 2023, and its repeal for solar and wind was announced in summer 2025. At the same time, it takes several years for a new generator project to go from inception to generating electricity on the national grid. Therefore, the short-run effects of a new subsidy policy may differ from the long-run effects. In the short run, over the first few years of a new subsidy policy, the subsidy can only affect the late stages of previously planned generators' development processes. In the long run, a subsidy policy can cause newly planned generators, as well as affect their development processes.

I measure the short-run effect of the Energy Community Bonus on new generator construction during its few years of enactment. Notwithstanding the subsidy repeal, I also measure the long-run effect of the subsidy on generator construction as if it were left in place through the mid-2030s as intended. I do this by exploiting the institutional structure that governs new generator planning and development. That planning process begins with a prospective project developer choosing where to locate a potential new generator and entering an "interconnection queue." The interconnection queue is a multi-year process by which projects learn how much it will cost to connect to the national grid. These costs are highly uncertain before the interconnection queue process; about 80 percent of queued projects ultimately reject the costs and are canceled. I measure the long-run subsidy effect by observing how the Energy Community Bonus affects queue entry and the probability projects accept their interconnection costs during the few years of the policy's enactment.[2,3] Given the subsidy is place-based, it is clearly important to measure an effect that accounts for generator location choice, which is essentially where generators enter the interconnection queue. Furthermore, it is important to estimate this long-run effect because the interconnection queue features inter-project externalities that could limit or bolster subsidy effects, such that the short- and long-run effects could qualitatively differ.

In the short-run, from 2023 through 2025, the Energy Community Bonus increased total clean electricity capacity in Energy Communities by 10 to 21 gigawatts (GW) per year. That means about a quarter of US grid capacity additions during that time were due to the Energy Commu-

[2]This is closely related to the idea of using "surrogate" short-run data to measure long-run treatment effects, described in Athey et al. (2026). Their method uses a reduced-form strategy to develop a "surrogate index" to measure long-run treatment effects from short-run surrogate data. Instead, I employ a framework that takes a structural stance on how short-run queue entry and cost acceptance probability can be combined to calculate long-run capacity additions.

[3]In the sense of the traditional microeconomic definition of the short run and long run, in which some factors of production are fixed in the short run and variable in the long run, the short-run effect holds queue entry fixed, while the long-run effect allows it to vary. I sharpen this in Section 2

nity Bonus. The long-run subsidy effects implied by its effects on interconnection queue entry and project survival probability are even larger. The subsidy nearly doubles interconnection queue entry in Energy Communities. Moreover, it causes a higher probability that projects accept their interconnection costs. I show the underlying mechanism is that the subsidy capitalizes into projects' willingness to accept (WTA) for their interconnection costs. Taken together, the subsidy more than doubles capacity additions in Energy Communities in the long run, causing a national increase in clean electricity capacity between 23 GW and 47 GW per year.

One concern with using spatial variation to measure subsidy effects, underscored by the large results, is the possibility that empirical estimates are biased by spillover effects. Specifically, the empirical strategies in this paper all compare outcomes in Energy Communities to outcomes in comparable areas that do not receive the subsidy. But if subsidizing projects in Energy Communities causes less clean electricity capacity elsewhere, then such an empirical strategy would overstate subsidy effects. To address this, I show empirically that, if anything, the Energy Community Bonus *crowds in* clean electricity in neighboring areas that do not receive the Bonus. In more distant areas, I show using a simple theoretical model that any spillover effects are likely trivially small. While the empirical evidence is only suggestive, it indicates that the results, large as they may seem, could be conservative estimates.

Physical effect estimates in hand, I turn to the emissions and welfare effects of clean electricity subsidies. The fiscal cost per ton of carbon dioxide ($CO_2$) abated for a marginal subsidy change is \$29 per ton for solar, \$33 per ton for wind, and \$106 per ton for batteries. I measure the marginal welfare effects of an increase in clean electricity subsidies using the marginal value of public funds (MVPF) (Hendren and Sprung-Keyser, 2020). The MVPF is a benefit-cost ratio: It is society's willingness to pay (WTP) for a marginal subsidy increase per fiscal dollar expended on that increase. Across all technologies, I find a large MVPF for clean electricity subsidies, about 6.5. There is substantial technology heterogeneity, with an MVPF for solar of about 9, for wind 7, and for batteries 3. The less favorable battery values are due to the emissions from charging batteries. Nonetheless, the battery MVPF is still quite high because the electricity consumed when batteries charge tends to be cleaner than the electricity displaced when they discharge. To some extent, the subsidy effect estimates drive the large MVPFs. Even more so, the large results reflect a large wedge between the clean electricity externality and the prevailing subsidy level.

I take two strategies to evaluate whether the Energy Community Bonus improves or worsens efficiency. First, I compare the MVPF of a subsidy to Energy Communities to the MVPF of a subsidy to the full US or a randomly chosen set of counties. Second, I use a sufficient statistics framework to calculate the marginal welfare effects of the Energy Community Bonus relative

to a capacity-neutral policy, one that results in the same amount of electricity capacity as the place-based policy, but offers a uniform subsidy across the country. The Energy Community Bonus raises the solar MVPF by about 0.2 and the wind MVPF by 0.3, so the distributional objective of targeting Energy Communities also improves policy efficiency. On the other hand, the place-based incentive lowers the battery MVPF by about 0.05. Compared to the total MVPF magnitudes, these changes seem small, but the sufficient statistics model shows they imply large welfare effects. For every year of subsidy disbursed, the Energy Community Bonus for solar and wind increases global welfare by $800 million in present value. The welfare loss for batteries is $1.1 billion per year. So, on net, the place-based policy poses a trade-off between the distributional objective and welfare.

The battery welfare loss outweighs the combined solar and wind welfare gain despite the smaller MVPF difference because batteries account for a large share of future capacity additions. Solar and wind subsidies are not more effective in Energy Communities because these areas are sunnier or windier. In fact, the opposite is true. Instead, the Bonus increases the welfare effects of solar and wind subsidies because, on average, a marginal generator in an Energy Community displaces dirtier electricity than a marginal generator elsewhere. This mechanism does not imply the same unambiguous efficiency improvement for batteries. It turns out that a battery in an Energy Community is, on average, more emissions-intensive than a battery elsewhere, so the place-based provision lowers the social value of battery subsidies.

**Related Literature** I contribute to the empirical economic research on clean electricity subsidies in the US. Previous work compares output and investment subsidies, showing that wind turbines receiving output subsidies generate more electricity than those receiving construction subsidies (Aldy et al., 2023). Johnston (2019) shows that making the clean electricity subsidies refundable increases their value to the subsidy recipients, especially non-incumbent recipients with less tax liability. Hitaj (2013) and Metcalf (2010) are early studies testing whether US clean electricity subsidies cause more wind capacity. I revisit the wind capacity effects of these papers, exploiting quasi-experimental variation, and contribute estimates for solar and battery storage. Hahn et al. (2026) use these estimates of wind elasticities to generate estimates of the MVPF for wind. I update the wind results using contemporary elasticities, contribute new estimates of the MVPF of solar and batteries, and investigate how the place-based policy changes these results.

I also contribute to the research on efficiency-equity trade-offs in environmental policy. In the case of carbon pricing, it is clear that incorporating distributional mechanisms worsens efficiency (Goulder et al., 2019). In the case of dirty energy, it is clear that a subsidy entails a welfare loss (Hahn and Metcalfe, 2021). In those contexts, the crucial question is at what welfare cost.

Conversely, in the case of subsidies for clean goods, that welfare cost can be positive or negative because these subsidies are almost never priced proportionately to the externalities the clean goods generate.[4] Linn (2022) shows that means-testing a subsidy for electric vehicles improves the subsidy cost-effectiveness because lower-income households are less likely to be inframarginal buyers.

Finally, I contribute to a new and growing literature on the Energy Community Bonus. Balkir (2026) and Keuzenkamp et al. (2026) also estimate the Energy Community Bonus effects on new capacity additions during the early years of the policy, also finding a positive result, in addition to estimating employment and political election effects. Zhang (2025) employs a calibrated economic model for a forward-looking perspective on Energy Community Bonus effects. Doshi et al. (2026) estimate the Bonus effects on interconnection queue entry, which is part of the long-run effect that I explore in this paper. They also use a structural model of generator location choice to model various subsidy designs. For the short-run outcomes on new generators, I contribute a different empirical strategy. I use a simple difference-in-differences (DID), as well as a synthetic DID (SDID), to estimate physical subsidy effects as transparently as possible. I also include a forward-looking interconnection queue analysis looking not only at queue entry but also at how projects fare after entering the queue. As I explain below, the subsidy effect on post-queue-entry dynamics is theoretically ambiguous, and proves to be quantitatively important. Lastly, I differ by measuring the emissions and welfare effects of the place-based policy and clean electricity subsidies generally.

I next describe the Energy Community policy and the process of building clean electricity generators in more detail. Section 2 presents a model of how clean electricity subsidies affect capacity in the short and long runs. Section 3 describes the data, Section 4 presents the empirical strategy and results, Section 5 shows the welfare and emissions model and results, and Section 6 concludes.

[4]By clean subsidies, I refer to subsidies for substitutes for emissions-generating goods, as opposed to carbon abatement subsidy, like direct air capture or carbon capture and sequestration. In the case of a direct price for carbon abatement, a uniform subsidy is cost-effective, so a place-based incentive would necessarily pose an efficiency-equity trade-off. Another case with differing intuition is that studied by Fullerton and Stechuk (2026), in which policymakers cannot price the most emissions-intensive good (e.g., coal), but can price less emissions-intensive goods (e.g., natural gas and renewables). In their case, it is not clear that the second-best optimal clean subsidy is even positive.

# 1 Background

## 1.1 Clean Electricity Subsidies and the Energy Community Bonus

The US federal government has long offered two nationwide tax credits for clean electricity.[5] The Investment Tax Credit (ITC), enacted with the Energy Tax Act of 1978, subsidizes construction costs for new clean generators. The Production Tax Credit (PTC), placed in force by the Energy Policy Act of 1992, subsidizes 10 years' worth of electricity generation on a per kilowatt-hour (kWh) basis. The credits cannot be combined, so electricity project developers must choose which to claim. The IRA set the ITC to 30 percent of construction costs, and the PTC to 0.3 cents per kWh in 1992 value, adjusted to inflation (0.6 cents per kWh in 2026) (CRS, 2026).[6] The Energy Community Bonus is a larger increase in the subsidy value for the ITC than for the PTC. It increases the ITC by 10 percentage points (30 to 40 percent) and the PTC by 10 percent (0.6 to 0.66 cents per kWh in 2026).

Table 1 lists the three types of definitions of Energy Communities, and Figure 1 shows the qualifying locations.[7] An area only needs to meet one of these definitions to be an Energy Community, so the full set of areas receiving the subsidy is the union, not the intersection, of these definitions.

| *Category* | *Spatial unit* | *Criteria* |
|---|---|---|
| Fossil fuel employment and unemployment | Groups of counties | Fossil fuel industry employment higher than 0.17 percent at any time after December 31, 2009 and a local unemployment rate higher than the national unemployment rate in the previous year. |
| Coal closure | Census tracts (subdivisions of counties) | Contains a coal mine closed after December 31, 1999 or a coal-fired electricity generating unit retired after December 31, 2009, or neighbors such a tract. |
| Brownfields | Parcels (subdivisions of census tracts) | Brownfield sites, based on a definition specific to the Energy Community Bonus. |

Table 1: Energy Community definitions.

[5] Historically through 2024, only certain clean electricity technologies were eligible for each tax credit. Technologies that are eligible for both the ITC and PTC include solar, wind, municipal solid waste, and geothermal electric. Only eligible for the ITC are energy storage, fuel cells, combined heat and power, microturbines, microgrid controllers, and interconnection property. Only eligible for the PTC are hydroelectric, biomass, and landfill gas. Beginning in 2025, due to the IRA, the tax credits became "technology neutral," such that any clean electricity is eligible to claim the credits regardless of technology.

[6] These values assume the project meets wage and apprenticeship requirements for 10 years. They exclude the value of the possible domestic content bonus, which is worth the same as the Energy Community Bonus.

[7] According to the IRA law text, there is another type of Energy Community not listed in this table: areas with sufficiently high local tax revenue coming from fossil fuel industries. However, Treasury has never identified the data defining local tax revenues, so the policy is not in effect and its definition is moot.

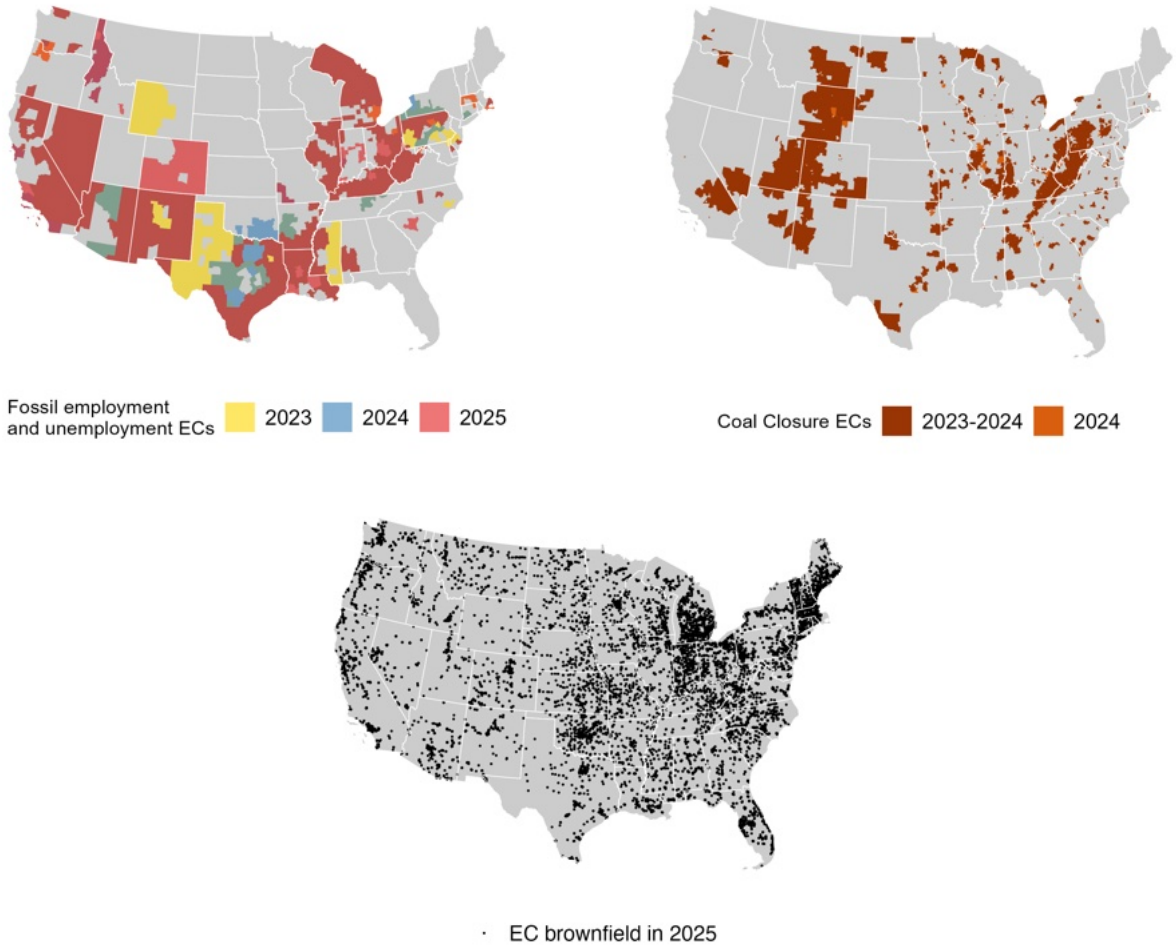



Figure 1: Energy Community (EC) locations. Brownfields are shown as point locations and are not to scale.

The first definition in the table (top-left map) covers areas with a sufficiently high historic fossil fuel employment rate and a high previous-year unemployment rate. According to this definition, Energy Communities can and do change year to year, as local unemployment rates and the national unemployment rate change. Many counties have this designation for at least one year, especially in the Southwest, Texas, and the Rust Belt. Fewer counties have the designation in all three years (shown in dark red, the combination of the semi-transparent colors). A project retains permanent Energy Community status if its location is designated as an Energy Community when it begins construction.

The other two Energy Community definitions—coal census tracts and brownfield parcels—are permanent designations. Coal Energy Communities are areas that have had a historic coal mine or coal-fired electricity closure. They are scattered throughout the eastern half of the US, and

cover large swaths of Appalachia and the Mountain West. Brownfields are polluted parcels of land, typically from former industrial use. While the Brownfields map in Figure 1 has much more colored area than the other two maps, the brownfields underlying the colored areas are small, as opposed to the other two maps, where the entire colored areas are Energy Communities.

According to the IRA law, the Energy Community Bonus, along with all clean electricity subsidies, was set to phase out in 2034 or when the US cut emissions 75 percent below 2005 levels, whichever happens first. However, the OBBBA accelerates that timeline for solar and wind technologies.

### 1.1.1 Partial Subsidy Repeal

The OBBBA, announced July 2025, two and a half years after the Energy Community Bonus enactment, accelerates the phase-out of the ITC, PTC, and Energy Community Bonus for solar and wind technologies. Solar and wind projects must begin construction before July 4, 2026 to be eligible for the tax credits.[8] The ITC and PTC phase-outs are unchanged for battery storage, hydroelectricity, geothermal, nuclear, and all other clean technologies. These phase-out deadlines only apply to new projects. For example, a project that claims the PTC before the policy lapses is guaranteed the full 10 years of subsidy, even after the policy lapses for new projects.

## 1.2 Building a Clean Electricity Generator in the US

Roughly, building a new clean electricity generator happens in three stages over several years. First, the project is planned. The project developer chooses a location, technology type, and project size. Second, the project enters an interconnection queue, and learns how much it will cost to connect to the national electricity grid. Finally, should the project developer choose to accept its costs, it finishes acquiring the necessary permits and financing, constructs the generators and necessary infrastructure, and is placed in service.

Interconnection queues are managed by local grid operators, each with its own set of rules and management systems, in addition to overarching rules from the Federal Energy Regulatory Commission (FERC). One common feature across the various interconnection queues is that it is not especially costly to enter an interconnection queue, at least relative to other costs of generator development and expected future revenue upon being placed in service. Entry costs are measured in tens of thousands of dollars, while interconnection costs are routinely tens of

[8] A project is also eligible if construction begins after July 4, 2026 and is placed in service by December 31, 2027, but that is widely considered an unrealistically fast turnaround from beginning construction to operation.

millions.[9]

While a project is on the interconnection queue, it receives a series of increasingly accurate interconnection cost estimates, based on "interconnection cost studies" conducted by the grid operator. Grid operators are regulated entities with no profit incentive. The cost study results are simply engineering estimates of the costs required to accommodate the new project, such as new and upgraded transmission lines, substations, and the labor required to build this infrastructure. After each cost study, a project has a fixed time window, usually about a month, to accept or reject the costs.

A project's interconnection cost is highly uncertain before it undergoes this series of studies because a project's infrastructure requirements and resultant costs depend on what other projects enter the queue, and how those projects react to their estimated costs. If another project withdraws from the queue after a cost study, that could lower costs for other projects because less ambitious grid upgrades are required, or it could raise costs for other projects, as grid upgrade costs are split among fewer projects. This is one reason why there are multiple cost studies. Projects withdrawing from the queue could trigger a re-study to update the cost estimates for the remaining projects. The other reason for successive studies is cost-saving, as the later studies are more complicated and expensive, but also more accurate.

Given the inter-project withdrawal externalities, grid operators impose penalties for withdrawal, which increase as the project advances through the queue process. In late stages, the penalties can reach millions of dollars. However, large study-to-study cost estimate changes are common. To limit project developers' risk of exposure to extreme cost estimate changes, grid operators often include caveats such as no withdrawal penalty if interconnection costs more than double.[10]

[9]To enter the queue, a developer just needs to demonstrate land ownership or an option to purchase land, and/or pay a mostly-refundable deposit. The rules vary by local grid operator. For example, the California Independent System Operator (CAISO) allows entrants to pay a $500 thousand to $2 million refundable deposit instead of demonstrating land control (CAISO, 2024). Midcontinent Independent System Operator (MISO) charges a $7 thousand application fee, plus $8 thousand per MW for the interconnection study (MISO, 2026). PJM requires generators to pay a deposit ranging from $10 thousand to over $300 thousand depending on the generator size, but only 10 percent of this deposit is not refunded if a project is withdrawn early in the queue process. Otherwise, the deposit is returned less 10 percent and the study costs, which are tens of thousands of dollars (PJM, 2026).

[10]For example, for ISO New England (ISO-NE), after the first study the penalty is two times the cost to conduct the study (tens of thousands of dollars), 5 percent of estimated network upgrade costs after subsequent studies (the "cluster re-study" phase), 10 percent of estimated network upgrade costs after a late-stage study (the "facility studies" phase), and 20 percent of estimated network upgrade costs if a project withdraws after it signs an interconnection agreement. If the cost estimate increases by over 25 percent in an early study, or if the estimate more than doubles after a later study, then there is no withdrawal penalty (ISO-NE, 2025).

After a project completes its cost studies, it signs an "interconnection agreement" to officially agree to its costs.

Appendix Figure A.1 shows the interconnection queue process as a game tree. I next define the short- and long-run subsidy effects and how to exploit the interconnection queue process to estimate the long-run effect of the Energy Community Bonus despite the policy repeal.

# 2 Model of Short- and Long-Run Subsidy Effects

This section presents a simple model to sharpen how I define subsidy effects in the short and long run, and to highlight an assumption required to calculate long-run capacity additions. I estimate the components of the model and calculate long-run subsidy effects in Section 4.

Consider capacity additions $x_j$ in some county $j$. These capacity additions are determined by the interconnection queue process, which unfolds over two time periods. In $t = 1$, projects choose whether to enter the queue. In $t = 2$, those projects learn and choose whether to accept their interconnection costs.

Capacity additions $x_j$ are determined by the capacity entering the interconnection queue times the probability those entrants accept their costs and "survive" the queue to become operational:

$$x_j(D_1, D_2) = e_j(D_1)\rho_j(e_j(D_1), D_2),$$

where $e_j$ is queue entry, and $\rho_j$ is the probability that queue entrants survive the queue. Queue entry and survival probability depend on $D_t$, which indicate whether the generators believe they receive the Energy Community Bonus upon becoming operational.

The number of queue entrants $e_j(\cdot)$ is a function of $D_1$, an indicator equal to 1 if projects believe they will receive the Energy Community Bonus when they make their queue entry decision. Those entrants' probability of becoming operational $\rho_j(\cdot, \cdot)$ depends on the set of queue entrants $e_j$ and $D_2$, an indicator equal to 1 if projects believe they receive the Bonus as they make decisions that determine whether they survive and become operational.

In the short run, the subsidy affects the probability of survival for a cohort of queue entrants that are incumbent in the queue as the subsidy is enacted. These are projects whose process is interrupted with news of the subsidy, and therefore only consider it in late stages of the process. The short-run treatment effect is

$$\tau_{SR} = x_j(0, 1) - x_j(0, 0).$$

This outcome is measurable by using data on new generators added to the electricity grid from 2023 to 2025. The Energy Community Bonus may have affected these new generators during the late stages of their development processes, but almost surely did not affect their queue entry decision.

The long-run subsidy effect on capacity additions endogenizes its effect on both queue entry and survival probability conditional on that queue entry,

$$\tau_{LR} = x_j(1,1) - x_j(0,0)$$

This is the treatment effect that a policymaker would expect if it enacts a subsidy and leaves it in place indefinitely. The challenge is that, because the subsidy was repealed after only a few years, $x_j(1,1)$ is not directly observable using data on new generators.

Instead, I measure $\tau_{LR}$ by assessing how the subsidy affects queue entry $e_j(\cdot)$ and survival probabilities $\rho(\cdot,\cdot)$:

$$\tau_{LR} = e_j(1)\rho_j(e_j(1),1) - e_j(0)\rho_j(e_j(0),0). \tag{1}$$

It turns out that queue entry and survival probability are resolved years before generators begin operation. That lag is caused by permitting and construction. Therefore, I can observe $e_j(1)$ and $\rho_j(e_j(1),1)$, and thereby $x_j(1,1)$, without needing to observe new generators coming online and despite the subsidy repeal.

In particular, I measure $\tau_{LR}$ by calculating the treatment effects

$$\begin{aligned} \tau_e &= e_j(1) - e_j(0), \\ \tau_\rho &= \rho_j(e_j(1),1) - \rho_j(e_j(0),0). \end{aligned}$$

I measure these treatment effects using reduced form identification strategies, and describe the identifying assumptions required to identify $\tau_e$ and $\tau_\rho$ as I present the analyses.

An additional key assumption for the long-run measurement exercise is that generators behaved as if they believed the solar and wind subsidies and Energy Community Bonus would remain in force indefinitely. That is, the subsidy repeal was a surprise, and $e_j(1)$ and $\rho_j(e_j(1),1)$ are therefore observable. I argue that this is a plausible assumption.[11] Qualitative evidence from late 2024 and early 2025 suggests the repeal of the ITC and PTC for solar and wind was widely considered unlikely. The clean electricity subsidies had existed for decades, surviving through many Democratic and Republican administrations, including the first Trump administration.

[11] I rely on qualitative argument, as I am not aware of any sufficiently thick prediction markets to quantitatively assess this probability (Wolfers and Zitzewitz, 2004; Meng, 2017).

Indeed, an often-cited justification for the IRA's subsidies over other possible climate and energy policies was its perceived political durability. From the perspective of 2023 through early 2025, repealing the clean electricity subsidies would require an act of legislation, which would at minimum require a Republican "trifecta," a scenario in which the US Republican Party wins a series of elections in November 2024. One outlet predicted a joint probability of a Republican trifecta of about 23 percent on election night.[12] This was a necessary but still insufficient condition for the policy repeal. Even after the November 2024 election, when a Republican trifecta materialized, it was still highly uncertain what would come of the ITC, PTC, and Energy Community Bonus. Many Republican congressmembers favored the clean electricity subsidies (Politico, 2024). In January 2025, experts speculated that the clean electricity subsidies were among the least likely Inflation Reduction Act subsidies to be repealed (Fujii-Rajani and Patnaik, 2025). Thus, I argue the repeal of the PTC, ITC, and Energy Community Bonus was perceived as improbable until June 2025, and project developers in interconnection queues believed they would receive subsidies until then.

I next describe the data I use to estimate the short- and long-run effects of clean electricity subsidies, and then turn to empirics.

# 3 Data

**New Electricity Generators** New capacity addition data are from Energy Information Administration (EIA) 860M forms. The data include the generator's technology type, capacity in MW, and point location. The sample begins in 2017 and ends in 2025.

**Interconnection Queue Entry** I obtain queue entry data from Interconnection.FYI.[13] The data include project-level observations of queue entrants, including their technology type and MW capacity. Unlike the point-located EIA generators, most queued projects are only spatially precise to the county level. The queue data cover all seven major ISOs and RTOs, plus 49 non-ISO utilities, accounting for about 97 percent of US electricity generation (Rand et al., 2025). I aggregate the data to the county level since this is the most precise spatial unit that covers all available areas. I drop some areas due to insufficient queue entry data, shown in Appendix Figures B.1 and B.2. Most notably, I drop the Electric Reliability Council of Texas (ERCOT), which covers about 8 percent of the US counties, and about 12 percent of US generating capacity. The sample begins in 2017 and goes through June 2025, just before the repeal announcement.

[12] 50 percent chance of Republicans winning the presidency, 49 percent the House, and 92 percent the Senate, according to 538 on November 5th, and assuming independence of the events (538, n.d.).

[13] Interconnection.FYI provides the data for the popular Queued Up report from the Lawrence Berkeley National Laboratory (LBNL) (Rand et al., 2025).

**Interconnection Cost Studies** Interconnection cost study data are available for Southwest Power Pool (SPP), MISO, and PJM. Other grid operators have some study data available, but none have sufficient historic data from before and after 2023. Interconnection.FYI provides the SPP and MISO data. PJM data are from a combination of Interconnection.FYI, Lawrence Berkeley National Laboratory (LBNL), and my own web-scraping from public sources. For a sense of the scope of this coverage, Appendix Figure B.3 shows each grid operator's coverage. Fortunately, these ISOs cover many Energy Communities and non-Energy Communities. The cost data are at the study-level, recording the study number (or phase), the date of the study, and the quoted interconnection cost. I merge this to the Interconnection.FYI data, including withdrawal dates for projects that withdraw, to infer whether projects accept or reject their estimated interconnection costs.

**Energy Communities** Fossil fuel employment thresholds are defined using US Census County Business Patterns (CBP) for 2010 to 2024 (Census, 2025). Local unemployment rates are calculated using the Bureau of Labor Statistics (BLS) Local Area Unemployment Statistics (LAUS) for years 2022 and 2023, to define Energy Communities in years 2023 and 2024, respectively (BLS, n.d.). I identify census tracts with coal mine closures and coal generator closures using the official data from the US Department of Energy (DOE) for tax years 2023 and 2024 (DOE, 2023; DOE, 2024).

There is no publicly available data source on the brownfields that meet the Internal Revenue Service (IRS) Energy Community Brownfield definition, but there is public data on Environmental Protection Agency (EPA) Brownfields from the EPA Facility Registry Service (EPA, 2025). Energy Community brownfields are a subset of the EPA brownfields (Service, 2025). I use the EPA data as the Energy Community brownfield set, dropping some observations that are unlikely to qualify for the Energy Community Bonus based on the law text (more on this in the Appendix).

**Socioeconomic and Geographic Data** I use various data sources to get a sense of how Energy Communities compare to the wider US. The 2021 Census American Community Survey provides population estimates and the 2022 Quarterly Census of Employment and Wages average wages. Direct normal irradiance (DNI, or sunniness) is from the National Solar Radiation Database (NSRDB), summarized at the maximum pixel value in each county (Sengupta et al., 2018). Wind speeds are from the WIND Toolkit, and are summarized in the same way (Draxl et al., 2015). Transmission line data come from Our Grid Future, and 2020 land cover data from the National Land Cover Database (NLCD) via IPUMS NHGIS.

**Data for Welfare and Emissions Analyses** To calculate emissions and welfare effects, I

use a variety of data to model how a marginal unit of clean electricity affects emissions. Most of these data come from National Lab of the Rockies Cambium model (Gagnon et al., 2024). I also use solar and wind capacity factors from the Regional Energy Deployment System (ReEDS) model and technology costs from the Annual Technology Baseline (Sergi et al., 2024; NLR, 2024). The next section details the empirical analysis and results, which do not rely on these data, so I defer an in-depth description to after the empirical analysis, in Section 5.

More detail on data cleaning is in Appendix B.

# 4 Subsidy Effects on Electricity Capacity

This section presents the estimation strategies and results for the subsidy effects on clean electricity capacity. All the empirical analyses are difference-in-differences (DID) comparing outcomes in Energy Communities to outcomes in non-Energy Communities. Section 4.1 describes how I define Energy Communities and non-Energy Communities for the DID strategies. Section 4.2 presents the analyses of the subsidy effect on two outcomes, new capacity additions ($\tau_{SR} = x(0,1) - x(0,0)$) and queue entry ($\tau_e = e_j(1) - e_j(0)$). I present these analyses together because they are both panel data at the county level. Section 4.3 presents the analyses of subsidy effects on the probability that queued projects accept their interconnection costs ($\tau_\rho = \rho_j(e_j(1),1) - \rho_j(e_j(0),0)$) and projects' WTA for their interconnection costs (a mechanism). These are presented together because they use the same interconnection cost study dataset.

Section 4.4 explores the possibility of spillover effects that could bias the analyses. Specifically, increases in electricity capacity in subsidized areas could, in principle, change capacity in unsubsidized areas. This would violate the stable unit treatment value assumption (SUTVA) required for any causal strategy with a treatment and a control, including a DID. Finally, Section 4.5 combines the queue entry result from Section 4.2 and cost acceptance probability results from Section 4.3 to calculate the long-run subsidy effect on capacity additions ($\tau_{LR}$).

## 4.1 Identifying Energy Communities and non-Energy Communities for the Empirical Analyses

Particular care is warranted to define Energy Community "treatment" counties and non-Energy Community "control" counties. One reason is that queue data are county-level, while some Energy Community definitions are at the sub-county levels. Coal mine and coal generator Energy Communities are census tracts, and brownfields are parcels of land (Table 1). Therefore, there are some counties that are partially Energy Communities. It is not possible to tell if projects in

the queue data in those counties are eligible for the Energy Community Bonus. Another reason for care is that the fossil fuel employment definition of Energy Communities changes year-to-year. Project developers may not behave as if they receive the Energy Community Bonus in an area that is only occasionally an Energy Community because, from their perspective, it is uncertain that the location will be an Energy Community when the project begins construction. The last concern underlying treatment and control selection is the possibility of spillover effects.

The main specifications define Energy Communities as counties that (1) meet the fossil fuel employment designation in all years since the policy was enacted, or (2) are completely covered by coal mine or coal generator Energy Community census tracts. Non-Energy Community "control" counties are those that (1) have never met the fossil fuel employment designation, (2) have zero coal-eligible census tracts, (3) have fewer than ten brownfields, and (4) do not share a border with an Energy Community.

I assume that counties that are fossil fuel employment Energy Communities in all years from 2023 to 2025 are certainly Energy Communities from the project developers' perspective. Treatment counties include those fully covered by coal census tracts because it is certain that projects in those counties are in Energy Communities. Brownfields do not define treatment counties because there are no counties fully covered by brownfields.

I drop counties that contain any coal census tract from the control group given the uncertainty over whether projects in these counties receive the subsidy. It is impractical to drop all counties with a brownfield from the controls in the same way, since this would leave very few potential control counties. The cutoff of ten is arbitrary, but the results are insensitive to more or less restrictive cutoffs. If anything, including Brownfield-qualifying projects in the control group would bias the treatment effects toward zero because the control group receives a partial treatment effect, limiting the possible difference between the treatment and controls due to the subsidy.

One concern with using coal-fired electricity generator closures as a treatment assignment is that it could be endogenous with renewable electricity development. Perhaps the closure causes newly uncongested transmission lines or substations that make these areas good places to build renewables. Or, perhaps a closure signals a location is a particularly unprofitable one for electricity generation. I run robustness checks dropping this type of Energy Community from the treatment group, and the results are qualitatively similar. Coal mine closures do not carry the same concern because mines close due to reasons unrelated to renewable suitability, such as increasing mine costs and national trends in natural gas and electricity prices (Watson et al., 2023).

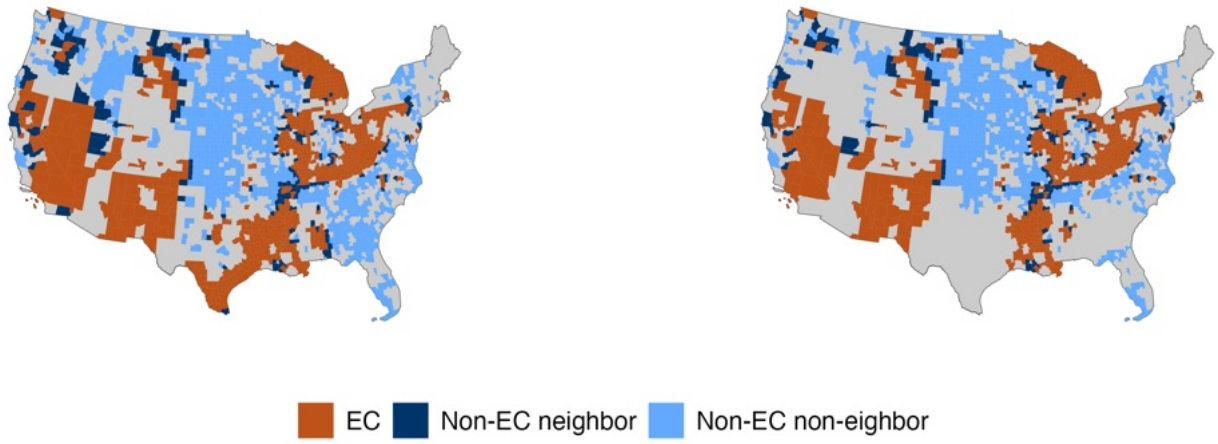
Lastly, I drop counties that neighbor Energy Communities from the set of potential control

counties for the empirical analyses. In Section 4.4, I explain why a SUTVA violation is more likely in neighboring counties, hence I drop them from the controls. This precaution turns out to be inconsequential. Including or excluding these neighbors from the control group does not qualitatively change results.

## 4.2 Subsidy Effects on Capacity Additions and Queue Entry

### 4.2.1 Descriptive Statistics

Table 2 presents descriptive statistics to compare the Energy Community county estimation sample to the rest of the US and the non-Energy Community control counties. Figure 2 shows the county locations. Given that Energy Communities are chosen to achieve distributional goals, it would be reasonable to suspect they are not a representative sample of US counties. However, Energy Communities are widely dispersed throughout the US, and turn out to be an approximately representative sample of the rest of the US, at least with respect to some key variables relevant to building a new electricity generator. Energy Community counties and the full US sample have similar populations, areas suitable for building clean electricity generators, and transmission wires. Energy Communities have somewhat more queue entry over the pre-period, but given the variance of the county-level aggregates, the values are similar. Energy Communities have had a similar level of capacity additions to the wider US in the pre-period.




(a) Capacity additions (b) Queue entry

Figure 2: Energy Community counties and controls. The difference between the treatment and control counties for capacity additions and queue entry is because the capacity addition data have full US coverage, while the queue data are missing certain areas.

The control counties have smaller populations and less transmission than the Energy Community counties, on average. They have similar levels of wind queue entry and capacity additions, but somewhat less solar and battery. There is a concentration of Energy Communities in the south and southwest, but fewer controls in this area. The treatment and potential controls also differ in their shares managed by each grid operator.

I next describe the strategies to estimate subsidy treatment effects on capacity additions and queue entry, which are designed to account for these differences between treatment and control counties.

### 4.2.2 Empirical Specification

I use two empirical specifications to estimate the short-run capacity addition ($\tau_{SR}$) and queue entry ($\tau_e$) effects, each with distinct advantages. The first is a two-way fixed effects DID with a grid-operator-by-year fixed effect. In particular, I estimate

$$Y_{jt} = \tau D_{jt} + \alpha_j + \gamma_{gt} + \varepsilon_{jt}, \tag{2}$$

where $Y_{jt}$ is either queue entry or capacity additions in county $j$ managed by grid operator $g$ in time $t$. $D_{jt}$ equals 1 if county $j$ is an Energy Community and time $t$ is after the Energy Community Bonus was enacted. $\tau$ is the average treatment effect on the treated (ATT).

The regression includes a grid-operator-by-year fixed effect.[14] This is to account for two potential threats to identification. First, grid operators have different speeds of processing queues. Second, grid operators collect queue entries in "clusters" of applicants, and have fixed schedules of processing clusters. If Energy Communities are more common in grid operators with faster queue processing or more post-period clusters, that would over-estimate the subsidy effects. The fixed effect accounts for potential spurious correlations between Energy Community prevalence and grid operator queue processing speed or cluster studies.

In event study specifications, I include all time periods, but in pooled treatment-effect estimation I drop some time periods surrounding the January 2023 policy enactment. In particular, I drop late 2022 from the pooled treatment estimates for queue entry analyses. The Energy Community Bonus was announced in August 2022, but not enacted until January 2023, and the guidance that detailed Energy Community locations was not announced until April 2023. Project developers likely speculated on the locations of Energy Communities before they were clear.

[14] Some counties are managed by more than one grid operator, so $\gamma_{gt}$ is not a fixed effect per se. It is a vector of dummy variables for being in the jurisdiction of grid operator $g$ in year $t$. I refer to it as a fixed effect here and elsewhere in the paper for simplicity.

| Variable | All Counties | Queue Entry Sample | | Cap. Add. Sample | |
|---|---|---|---|---|---|
| | | EC | Non-EC | EC | Non-EC |
| Sunniness (DNI, std. units) | 0 | -0.16 | -0.13 | -0.06 | -0.03 |
| | (1) | (1.33) | (0.69) | (1.28) | (0.72) |
| Wind at 100m (std. units) | 0 | -0.16 | 0.14 | -0.17 | 0.01 |
| | (1) | (0.97) | (0.86) | (0.94) | (0.93) |
| Population (10,000s) | 10.54 | 8.68 | 4.46 | 9.23 | 4.75 |
| | (33.56) | (49.4) | (11.5) | (49.39) | (11.3) |
| Area Developed ($mi^2$) | 31.79 | 27.88 | 18.02 | 30.29 | 18.56 |
| | (58.4) | (67.24) | (21.4) | (75.45) | (22.06) |
| Area Open or Ag. ($mi^2$) | 252.21 | 203.01 | 319.74 | 212.25 | 288.98 |
| | (229.96) | (204.76) | (245.35) | (204.72) | (242.4) |
| Area Bare, Grass, Shrub ($mi^2$) | 365.21 | 500.45 | 196.49 | 548.1 | 222.37 |
| | (1121.33) | (1607.14) | (470.83) | (1683.37) | (551.1) |
| Transmission (kV×km 1,000s) | 50.98 | 50.72 | 33.24 | 53.61 | 35.11 |
| | (74.26) | (110.59) | (34.75) | (107.73) | (39.01) |
| PJM or MISO | 0.51 | 0.8 | 0.58 | 0.72 | 0.47 |
| NYISO or ISONE | 0.05 | 0.01 | 0.04 | 0.01 | 0.03 |
| CAISO or SPP | 0.26 | 0.13 | 0.41 | 0.16 | 0.35 |
| ERCOT | 0.08 | 0 | 0 | 0.13 | 0.02 |
| Non-ISO | 0.49 | 0.39 | 0.47 | 0.38 | 0.55 |
| *Queue Entry 2017-2022 (MW)* | | | | | |
| Solar | 299.52 | 415.58 | 197.02 | | |
| | (697.23) | (864.71) | (455.43) | | |
| Wind | 149.81 | 105.24 | 136.38 | | |
| | (941.19) | (556.79) | (718.62) | | |
| Battery | 200.86 | 270.18 | 91.69 | | |
| | (825.11) | (1236.55) | (292.21) | | |
| Other Elig. | 3.75 | 4.22 | 2.24 | | |
| | (45.06) | (39.26) | (40.15) | | |
| *Capacity Additions 2017-2022 (MW)* | | | | | |
| Solar | 16.16 | | | 15.43 | 10.39 |
| | (79.24) | | | (111.58) | (48.83) |
| Wind | 19.42 | | | 19.39 | 19.37 |
| | (92.59) | | | (83.73) | (88.74) |
| Battery | 2.75 | | | 5.19 | 0.99 |
| | (37.06) | | | (65.42) | (19.57) |
| Other Elig. | 1.34 | | | 1.3 | 1.15 |
| | (14.37) | | | (12.12) | (17.6) |
| N | 3,108 | 621 | 1,024 | 739 | 1,261 |

Table 2: Descriptive statistics and balance table. Means with SDs in parentheses, except the grid operator measures, which are shares. Shares sum to more than 1 because some counties are covered by more than one grid operator. "All Counties" queue entry measures use the 2,558 counties with queue data. EC and non-EC counties sum to less than 3,108 due to dropping partially treated counties and, for the queue sample only, missing data. DNI and wind speed are measured as county maximums.

**Identification** $\tau$ in Equation 2 identifies the Energy Community Bonus effect on the average MW of capacity additions ($\tau_{SR}$) or queue entry ($\tau_e$) per county per year under the two standard DID assumptions. The first is no anticipation of treatment effects. This is why I drop periods surrounding the Energy Community Bonus enactment. I assume the Energy Community Bonus policy was not anticipated before summer 2022, and following the argument in Section 1, that the policy repeal was not anticipated. The second assumption is conditional parallel trends: that average outcomes in Energy Community counties within grid operator-years would have trended in parallel with the observed conditional outcomes in control counties if not for the Energy Community Bonus. The primary way I evaluate this is by testing whether trends are parallel in the pre-period. Notably, I do not assume that Energy Community assignment is random, nor do I assume that the levels of capacity additions or queue entry are equal in Energy Communities and controls.

The strengths of the DID specification in Equation 2 are its transparency and unbiasedness under the standard assumptions. However, the parallel trends assumption may be more plausible in some weighted subset of the control counties, as opposed to the equal-weighted full set of controls. The second empirical specification is a synthetic DID (SDID), which is estimated using an algorithm to select weights on control counties that have pre-trends that best match trends in the treated Energy Communities (Arkhangelsky et al., 2021). The SDID estimating equation is:

$$(\hat{\mu}, \hat{\alpha}, \hat{\eta}, \hat{\tau}) = \arg\min_{\mu,\alpha,\eta,\tau} \left\{ \sum_j \sum_t (\tilde{Y}_{jt} - \mu - \alpha_j - \eta_t - D_{jt}\tau)^2 \hat{\omega}_j \hat{\lambda}_t \right\}. \tag{3}$$

The equation is the same as the simple DID specification with a few key amendments.[15] First, the estimator chooses unit-weights $\hat{\omega}_j$ to minimize the difference between the pre-period trends between the treated and control units. Second, time-weights $\hat{\lambda}_t$ are chosen to select the pre-treatment periods that best match the post-treatment periods, such that the pre-period difference is not driven by outlier periods. $\mu$ is an intercept to allow for the parallel difference between the treatment and control trends.

The outcome variable is denoted $\tilde{Y}_{jt}$ with a tilde due to a technical detail that enables including the grid-operator-by-year fixed effect. I include the fixed effect following the method in Kranz (2022). In particular, I estimate the linear model

$$Y_{jt} = \gamma_{gt} + \delta_t + \rho_j + \varepsilon_{jt},$$

using all counties in the pre-period and only control counties in the post-period. This way, $\gamma_{gt}$ is the average effect on $Y_{jt}$ of being located in grid operator $g$'s jurisdiction in year $t$, not including

[15] I recycle some of the notation, such as $\alpha_j$, in Equations 2 and 3. This is an abuse of notation because the coefficients are not exactly the same, but they represent similar model features, so I denote them similarly.

the treatment effects. Finally, I calculate the adjusted $\tilde{Y}_{jt} = Y_{jt} - \gamma_{gt}$ for all treatment and control units. I bootstrap to include this first stage in calculating standard errors (Kranz, 2022). The SDID specification drops periods surrounding the policy enactment in the same way as the DID specification.

**Identification** The SDID algorithm estimates treatment effect $\tau$ as a weighted two-way fixed effect specification, so the identifying assumptions are similar to those required for DID. First, I assume no anticipation of the policy enactment or repeal during the sample periods. Instead of a parallel trends assumption, $\tau$ is identified assuming a weighted parallel trends assumption. The assumption is fundamentally untestable, but parallel trends in the pre-treatment periods support the assumption.

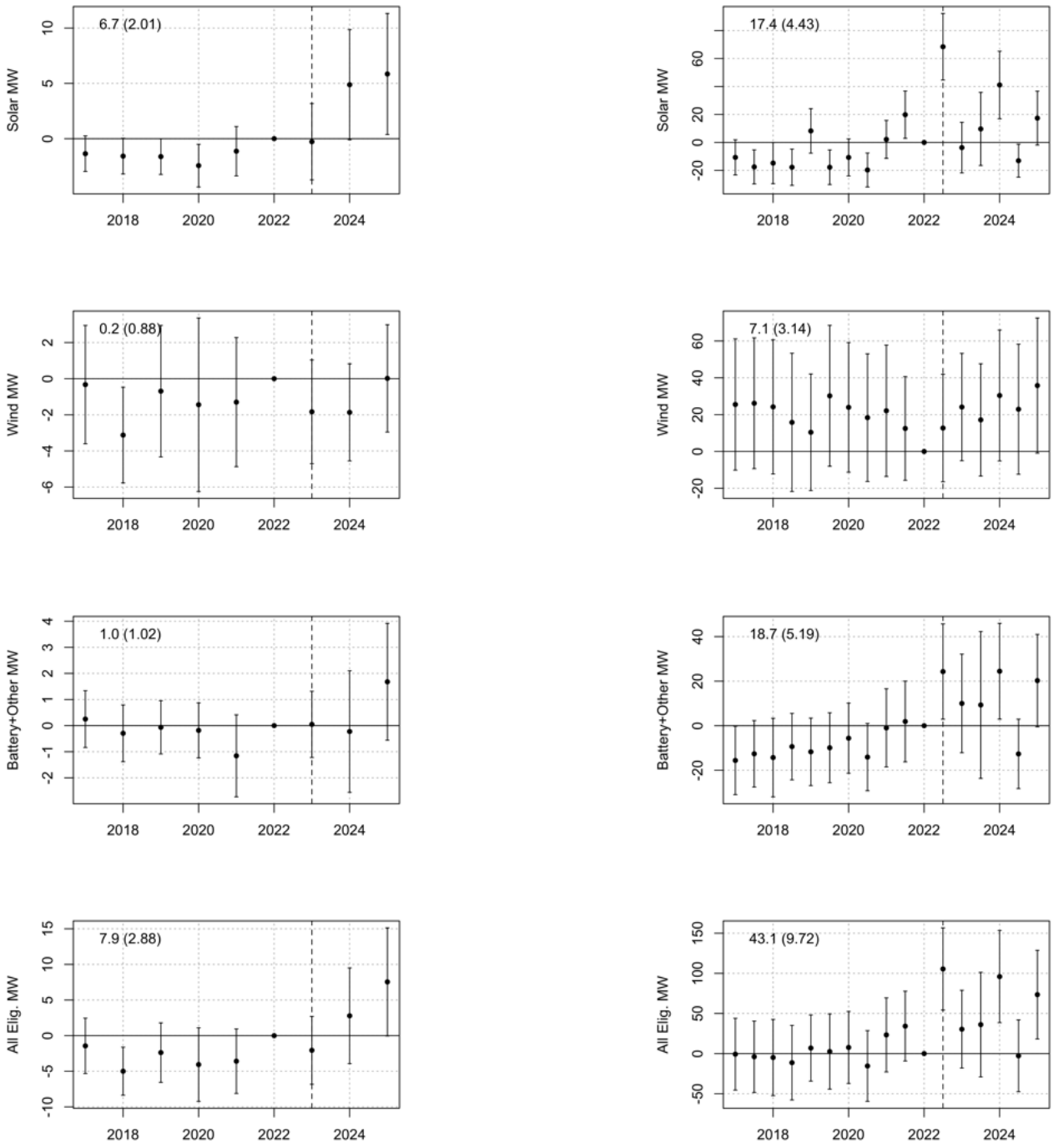
Another option for an empirical approach is to use a regression discontinuity design (RDD) on the fossil fuel employment definition of Energy Communities, which includes two sharp thresholds. However, RDD assumes the outcome variable is a continuous function of the running variable around the threshold. This is not the case with electricity capacity additions and fossil fuel employment rates, so the RDD produces imprecise results. While including many controls somewhat improves precision, this changes the interpretation of the estimand.

### 4.2.3 Queue Entry and Short-run Capacity Addition Results

The queue-entry regressions are at the half-year unit of analysis because the data end after the repeal announcement in July 2025, while the capacity addition regressions are at the annual level. The capacity addition regressions are measured in megawatts (MW) per county per year, and the queue entry regressions in MW per county per half-year.

Figure 3 shows the results of estimating the DID in Equation 2 as an event study, with pooled coefficients in the figure. The capacity addition event studies show parallel trends in the pre-period. The queue entry event studies show slight pre-trends, so the preferred specification is the SDID. However, any pre-trends are mild, and the estimated treatment effects are mostly qualitatively similar to those estimated by SDID.

Figure 4 reports the SDID results for queue entry and Figure 5 for capacity additions. The top row shows the mean trends for the Energy Communities and a synthetic control group. The bottom row shows a permutation test, generated by re-running the specification in Equation 3 1,000 times, and with each permutation re-sampling the counties with replacement and randomly assigning treatment status. The permutation tests are useful to get a sense of treatment effect size relative to the variation in the data.




(a) Capacity additions

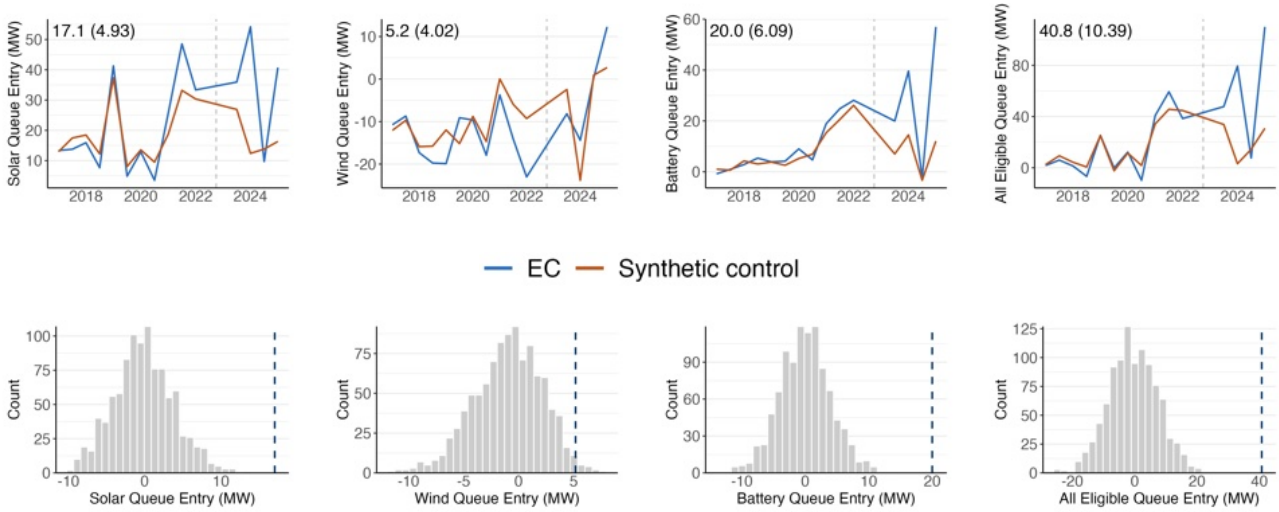
(b) Queue entry

Figure 3: Event studies, with grid-operator-by-year fixed effect. Pooled treatment effect at the top-left with standard errors clustered at the county level.

The SDID trend figures in the top row show the algorithm is effective in eliminating any pre-trends. For both queue entry and capacity addition outcomes, solar, battery, and pooled estimates are positive, much larger than would be expected under random treatment assignment, and statistically significant. Subsidy effects on wind are insignificant. The largest effects are for solar and batteries.

The effect sizes for the SDID are similar to the DID regressions, except for battery capacity additions, where they are larger. For a sense of magnitude of the capacity addition result, compared to a counterfactual without the treatment effect, the treatment effect suggests a doubling of solar capacity additions and a five-times multiplier on battery capacity additions. The queue entry results are a 70 percent increase for solar and more than a doubling for batteries.

For queue entry, most results show a drop in the difference effect between Energy Communities and non-Energy Communities in the second half of 2024. This is because it happens that the second half of 2024 did not have many interconnection queue request clusters.

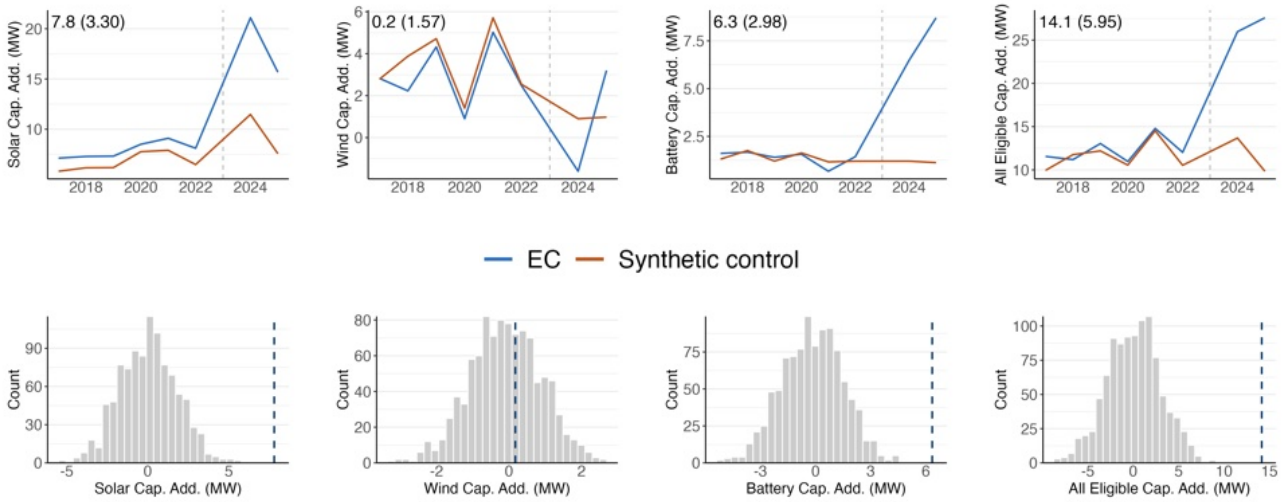



(a) Solar (b) Wind (c) Battery and other (d) Pooled

Figure 4: Queue entry SDID and permutation test results. Pooled treatment effects reported at the top, with bootstrapped standard errors in parentheses. EC stands for Energy Community.

### 4.2.4 Heterogeneity and Robustness Checks

Appendix C reports heterogeneity by grid operator. This exercise still requires some aggregation because certain grid operators do not have a sufficient number of Energy Community counties or non-Energy Community controls. The results tend to lose significance with the disaggregation, though some remain significant. MISO, SPP, CAISO, ERCOT, and the non-ISO regions have the same qualitative takeaways upon disaggregation. PJM, NYISO, and ISO-NE show null or




(a) Solar (b) Wind (c) Battery and other (d) Pooled

Figure 5: Capacity addition SDID and permutation test results. Pooled treatment effects reported at the top, with bootstrapped standard errors in parentheses. EC stands for Energy Community.

even insignificantly negative effects.

One concern highlighted by the maps showing Energy Community counties and non-Energy Community controls (Figure 2) is that the Energy Community counties tend to be in the sunnier west or southwest. The unweighted DID pre-trends being approximately parallel largely quells this concern for the main specification. But to further interrogate the result, Appendix D presents three more SDID analyses. The first drops Arizona, California, Nevada, and New Mexico. The second robustness check adds a state-by-year fixed effect to the baseline specification, which effectively drops Arizona, Nevada, and New Mexico because there is no variation in Energy Community status in those states' sampled counties. The third keeps the state-by-year fixed effect and also drops California.[16] Dropping the western states still yields positive and significant results for solar, batteries, and aggregate. The state fixed effect makes standard errors much larger, so some results lose significance. However, estimate magnitudes are similar, and some results even retain significance.

[16]I caveat that since states have varying numbers of Energy Community counties, including a state-by-year fixed effect eliminates a valid channel of treatment effect. If the subsidy has an effect, states with more Energy Communities should see larger average capacity addition or queue entry changes in the post-period, but the state-by-year fixed effect eliminates that channel. For this reason, the state-by-year fixed effect is an intentional omission from the primary specification. However, given all states have some capacity additions and queue entry in every year, it effectively controls for trends in geography.

Taken together, the results show the Energy Community Bonus has a substantial short-run effect on capacity additions. These results also show a precursor to a substantial long-run effect, a large effect on queue entry in Energy Communities due to the subsidy. However, more queue entry in Energy Communities does not alone guarantee that the subsidy would cause more capacity additions. I next investigate how the subsidy affects projects on the interconnection queue.

## 4.3 Subsidy Effects on Interconnection Studies and Costs

This section presents estimates of the effect of the Energy Community Bonus on the probability that projects accept their interconnection costs, or their queue "survival" probability, $\tau_\rho$. After estimating survival probability effects, I investigate one possible mechanism underlying the results: that the subsidy may capitalize into interconnection cost WTA. But first I describe the available data.

### 4.3.1 Descriptive Statistics

Table 3 presents descriptive statistics of the interconnection cost study data from MISO, SPP, and PJM. The left-most column shows the descriptives of the full dataset. The right two columns show Energy Community and non-Energy Community descriptives.

The mean quoted cost to connect to the grid from the first study is nearly $50 million, or $345 thousand per MW. The standard deviation is larger than the mean, underscoring the high variability in costs, and suggestive of the uncertainty. Non-first study costs are lower on average, reflecting the tendency of projects to reject studies with high costs. Even more suggestive of the cost uncertainty is that the standard deviation of the change in costs between studies is nearly as large as the mean cost.

The first three sections of Table 3 show the available data to measure interconnection queue survival estimates. Importantly, estimating the long-run effect in Equation 1 requires survival outcomes for projects that enter the queue after the Energy Community Bonus was enacted and before its repeal, such that the data yields $\rho_j(e_j(1), 1)$, not $\rho_j(e_j(0), 1)$ (survival of pre-Energy Community Bonus queue entrants).

There are hundreds of post-policy entrants and interconnection cost studies of post-policy entrants. However, many of those projects did not have an observable survival outcome pre-repeal. In other words, their outcomes are censored. In particular, 2024 and 2025 entrants and all entrants in PJM tend to be censored. The empirical specification below accounts for this censoring. The next section describes the specification in more detail.

| Variable | All Counties | All EC | Non-EC |
|---|---|---|---|
| N projects entering queue | | | |
| in pre-IRA | 6,934 | 2,252 | 1,628 |
| in 2023 | 874 | 215 | 232 |
| in 2023 (w.o. PJM) | 426 | 73 | 152 |
| in 2024-25 | 1,200 | 519 | 241 |
| N studies | | | |
| of pre-IRA entrants, pre-IRA | 3,069 | 852 | 889 |
| of pre-IRA entrants, 2023 or later | 3,872 | 1,305 | 1,047 |
| of 2023 entrants | 716 | 133 | 228 |
| of 2024-25 entrants | 58 | 23 | 13 |
| Share studied or withdrawn pre-study | | | |
| pre-IRA entrants | 0.85 | 0.85 | 0.87 |
| 2023 entrants | 0.49 | 0.33 | 0.63 |
| 2023 entrants (w.o. PJM) | 0.92 | 0.86 | 0.94 |
| 2024-25 entrants | 0.09 | 0.10 | 0.07 |
| Initial study cost ($M) | 49.2 | 61.8 | 41.2 |
| | (79.5) | (92.0) | (61.7) |
| Initial study cost per MW ($M) | 0.345 | 0.421 | 0.299 |
| | (0.539) | (0.614) | (0.439) |
| Non-first study cost ($M) | 41.0 | 57.0 | 31.0 |
| | (67.3) | (82.5) | (48.1) |
| Non-first study cost per MW ($M) | 0.251 | 0.344 | 0.208 |
| | (0.338) | (0.383) | (0.292) |
| Change from prior study ($M) | -1.1 | -1.5 | -1.0 |
| | (44.2) | (51.8) | (39.7) |
| Change from prior study per MW ($M) | -0.017 | -0.018 | -0.011 |
| | (0.334) | (0.307) | (0.307) |
| Share of withdrawn that withdrew pre-study or within 3 mo. of last study | 0.89 | 0.90 | 0.91 |
| Project technology shares | | | |
| Hybrid | 0.12 | 0.15 | 0.09 |
| Solar or wind | 0.70 | 0.70 | 0.72 |
| Battery | 0.19 | 0.14 | 0.19 |
| Capacity shares | | | |
| Battery | 0.20 | 0.19 | 0.18 |
| Solar | 0.60 | 0.70 | 0.51 |
| Wind | 0.19 | 0.11 | 0.31 |
| Project shares | | | |
| MISO | 0.47 | 0.64 | 0.33 |
| SPP | 0.24 | 0.11 | 0.37 |
| PJM | 0.29 | 0.26 | 0.29 |

Table 3: Summary statistics for the interconnection cost data. EC stands for Energy Community. Other than the counts and shares, the numbers are means with standard deviations in parentheses.

### 4.3.2 Interconnection Queue Survival Empirical Specification

The objective of the empirical analysis in this section is to estimate how the subsidy changes the probability that projects in Energy Communities accept their interconnection costs and become operational. I do this by measuring the complementary probability, that projects *withdraw* after their cost study.

I estimate this probability effect by estimating the equation,

$$W_{ist} = \tau_\rho D_{it} + \zeta EC_i + \gamma_{gt} + \varepsilon_{ist} \tag{4}$$

where $W_{ist}$ equals one if project $i$ managed by grid operator $g$ withdraws from the queue after study phase $s$ in time period $t$ and zero otherwise. $EC_i$ equals one if project $i$ is in an Energy Community and zero otherwise. $\tau_\rho$ is the change in the probability of project withdrawal due to the subsidy. Like the previous analyses, I include a grid-operator-by-year fixed effect.

Since projects receive multiple studies, or multiple study "phases," I estimate this equation twice for different study phases. The first iteration measures how the subsidy affects survival through the project's first study. The second measures how the subsidy affects survival through non-first studies. I use this separation instead of stacking all studies because projects often withdraw pre-study, and this separation avoids arbitrarily grouping pre-study withdrawal outcomes with study-level outcomes.

To limit censoring bias, I trim the post-period observations to 2023 queue entrants, and use only data for MISO and SPP, dropping PJM.[17] These two adjustments limit censoring, as about 90 percent of the rest of the post-policy entrants are uncensored, which is similar to the pre-policy censoring rate. In MISO and SPP, of the approximately 200 projects that entered the queue and had a study, nearly 99 percent of them have had a second study. Therefore, the non-first study outcome should also be little affected by censoring bias.

My main specification also drops projects and studies for projects that enter the queue before the policy, but do not have an outcome until after the policy. This ensures that the projects in Energy Communities exhibit survival probabilities conditional on queue entry accounting for the Bonus, $\rho(e_j(1), 1)$, and not $\rho(e_j(0), 1)$. I also run the analyses keeping these projects, with the caveat that the interpretation of the estimate is not a long-run effect.

[17] Another popular option to handle censoring bias is a duration model (Abbring and Van den Berg, 2003). However, these models would not debias the regressions because the censoring is a variable that projects can control, at least to an extent, as they select into receiving an early study or early withdrawal. Further, converting short-run duration model estimates to long-run survival would require a parametric assumption. Instead, I prefer the simple, unbiased, semi-parametric linear model. The economics literature on crime recidivism and unemployment policies takes a similar approach (e.g., Rose (2021) and Lichter and Schiprowski (2021)).

The downside of these approaches to censoring bias and estimand interpretation is that they limit statistical power. It turns out to yield sufficient power for analyses that aggregate all technology types, but proves overly limiting for certain technology-level regressions.

Perhaps notably, I do not control for interconnection costs. This is a bad control in that it is also affected by the subsidy. Controlling for interconnection costs would yield the withdrawal probability effect omitting whatever role is contributed by the higher costs shown in the previous section.

**Identification** This cross-sectional DID identifies $\tau_\rho$ assuming that the average probability of withdrawal for projects in Energy Communities would have trended in parallel to the average probability of withdrawal of non-Energy Community projects in the post-period, if not for the subsidy. To assess the plausibility of this assumption, I estimate Equation 4 as an event study to assess whether trends are parallel in the pre-period. The other necessary assumption is that project developers do not anticipate the policy enactment and repeal. I address this by dropping inframarginal queue entrants, which effectively drops observations of projects and studies that are near the policy enactment date, and I argue in Section 1 that the repeal was a surprise.

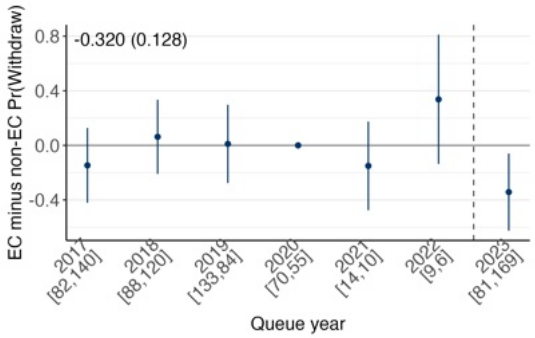
### 4.3.3 Interconnection Queue Survival Results

Figure 6 shows the results of estimating Equation 4 as an event study, with the pooled estimate and county-clustered standard errors at the top left. The left graphs show how the subsidy affects the probability projects withdraw pre-study or after the first study. The right graphs show how the subsidy affects withdrawal probabilities after non-first studies. The top row is the preferred specification, which drops projects that entered the queue pre-IRA but do not have outcomes until post-policy. The bottom row retains those projects in the estimating sample.

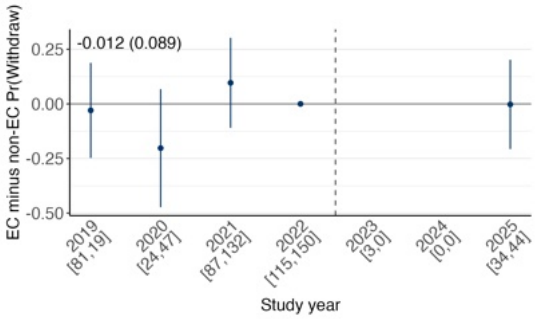
The top-left graph shows the Energy Community Bonus substantially reduces the probability projects withdraw, or equivalently, increases their survival probability. There is only one post-period, the 2023 queue entrants, because the 2024 and 2025 queue entrants are dropped due to censoring. Trends are parallel before the policy. The 2021 and 2022 periods are noisy because most of those pre-policy queue entrants do not have an outcome until post-policy, and are therefore dropped. Projects in Energy Communities after the policy are 32 percentage points less likely to withdraw (more likely to survive) before their study or after their first study.

The right panel shows the subsidy effect estimates on withdrawal after non-first studies. This regression is at the study level, unlike the previous one, which is at the project level. So, the x-axis is tracking the year of the non-first study. No projects in non-Energy Communities entered the queue in 2023 and had a non-first study before 2025, hence the lack of estimates. However,

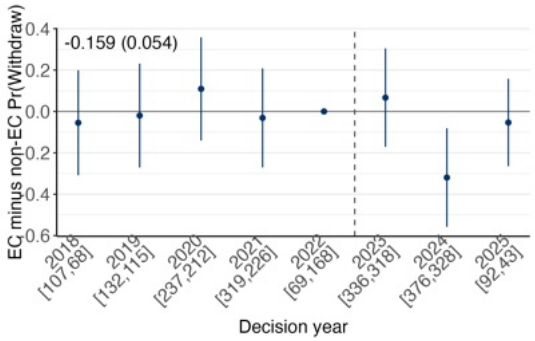
there are 80 non-first studies in 2025, and they suggest the subsidy has little or no effect on survival rates after the initial study period.



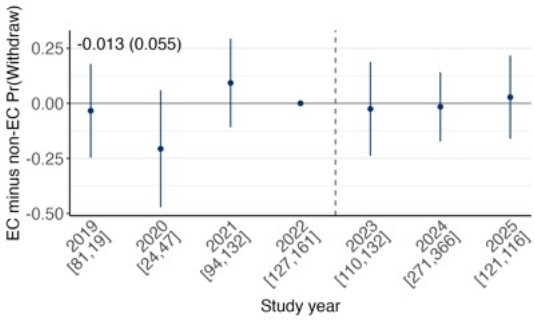

(a) Initial study, drop pre-IRA entrants




(b) Non-first studies, drop pre-IRA entrants




(c) Initial study, keep pre-IRA entrants




(d) Non-first studies, keep pre-IRA entrants

Figure 6: Event studies of how the subsidy affects projects' probability of withdrawing from the interconnection queue. Capacity-weighted. The pooled coefficient estimate is at the top left, with county-clustered standard errors in parentheses. The numbers in brackets below the years show the number of projects underlying the observation in Energy Communities and non-Energy Communities, respectively.

The bottom row of Figure 6 shows these same results, but retaining the inframarginal entrants. These analyses perhaps best serve to bolster the plausibility of the parallel trends assumptions. The time unit in panel 6c is the decision year. This is either the year that the project withdraws pre-study, or its initial study year. For the non-first studies, the time unit is simply the study year. Retaining the pre-IRA entrants yields results that are qualitatively similar to dropping those entrants.

I next investigate one mechanism that could drive the decline in withdrawal probability: that projects receiving the subsidy have higher WTA for their interconnection costs.

#### 4.3.4 Mechanism: Interconnection Cost Willingness to Accept

This section investigates WTA for interconnection costs, one mechanism by which the Energy Community Bonus could affect queue survival. The idea for the mechanism is simple: When projects receive their interconnection cost estimates from their cost studies, if they receive extra subsidy, their WTA for interconnection costs compared to similar projects should be higher by some amount up to the value of the subsidy. Not every project is the marginal project that will require the full value of the subsidy, so the difference in WTA between Energy Community and similar non-Energy Community projects should be less than or equal to the subsidy value.

Figure 7 shows the interconnection costs per MW for projects in MISO, SPP, and PJM. The leftmost panel shows the distribution of costs for withdrawn and not withdrawn projects, separated by technology.[18] Unsurprisingly, projects that have high costs tend to withdraw, and those that have low costs tend to not withdraw.

The middle and right panels show accepted costs by technology in Energy Communities and non-Energy Communities. The middle panel is pre-policy data, and the right panel is post-policy data. Before the policy, the mean costs in Energy Communities and non-Energy Communities are similar. After the policy, projects in Energy Communities accept higher costs. Appendix E estimates a DID and Figure E.1 shows the result as an event study with a pooled estimate. The annual cost data are noisy, but pre-trends are parallel and the DID has the same takeaway as Figure 7.

The DID yields a mean difference of about $142 thousand per MW; Energy Community projects are willing to accept about double the interconnection costs of non-Energy Community projects. To compare the difference in WTA to the subsidy value, in calculations later in the paper, I show the subsidy has a present discounted private value of about $145 thousand per MW for solar, $170 thousand per MW for batteries, and $80 thousand per MW for wind, with substantial variation by technology and location. This mean capitalization seems slightly high based on my own calculations, but it is within a sensible range and is evidence that the Bonus capitalizes into interconnection cost WTA.

Taken together, the previous analyses suggest large long-run effects of the Energy Community Bonus because projects are not only more likely to enter the queue in Energy Communities, but also more likely to survive the queue process and become operational. I calculate the implied long-run capacity addition effect in Section 4.5. Before this, I evaluate the possibility that subsidizing Energy Community projects affects projects in non-Energy Communities, which

[18] Withdrawing after a study is defined as withdrawing within 3 months of the study. Not withdrawing is defined as not withdrawing within 3 months.

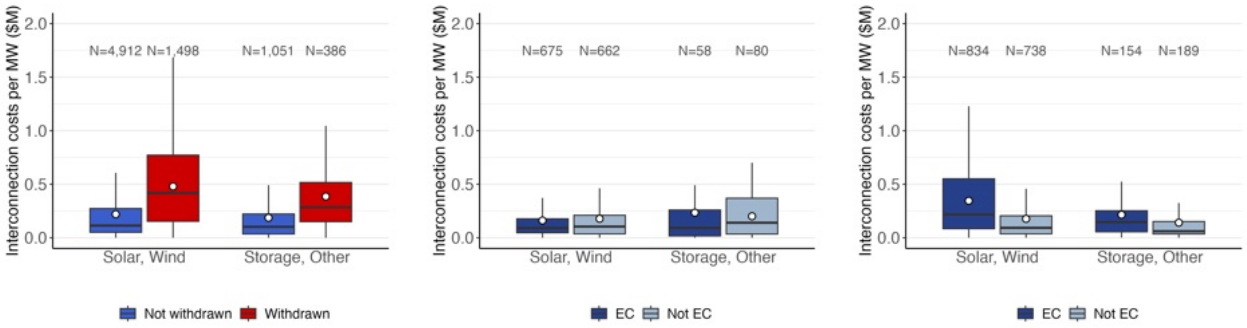



Figure 7: Interconnection costs per MW by technology and Energy Community (EC) type in real 2025 USD. Hybrid generation and storage projects are grouped with solar and wind. The boxes show the 25th and 75th percentiles, the black line the median, the white dot the mean, and the whiskers 1.5 times the interquartile range up to the minimum or maximum.

would bias the DID empirical strategies used in this paper.

## 4.4 Spillover Effects

A concern with any empirical strategy using spatial variation for identification is that the treatment affects the control group, violating SUTVA. In the context of this clean electricity subsidy, there are three plausible mechanisms that could facilitate a SUTVA violation. The first is about grid infrastructure costs, the second grid congestion, and the third traditional economic price mechanisms.

On infrastructure costs, building a new clean electricity generator may affect the cost to build other projects. This is an externality, and its value can be positive or negative. It can be positive because projects in the queue often must share the costs of grid infrastructure, so the presence of a project can lower costs for others. It can be negative because the presence of a project can increase grid infrastructure requirements for other projects. So, this mechanism could cause more or fewer projects in non-Energy Communities due to the subsidy.

The grid congestion spillover effect is another externality that could produce positive or negative spillovers. Building a new project could require building a new transmission line, which could lower grid congestion for other projects. Building a new project could also congest planned or existing lines, lowering the values of other projects. For batteries, grid congestion is a benefit, not a cost. Grid congestion lowers the marginal value of electricity on a transmission line to zero, creating an arbitrage opportunity. So, while batteries are still exposed to these externalities, the physical effects are associated with the opposite externality signs.

Finally, the traditional economic price mechanism is that subsidizing electricity in one area lowers the equilibrium price of electricity in all areas. Intuitively, the subsidy shifts the marginal cost curve downward, and lowers the price. This effect could materialize in markets for electric energy, as well as markets for electric capacity reliability.

### 4.4.1 Local Spatial Spillovers

I first investigate the net effect of these spillover mechanisms in counties nearby Energy Communities. Any of these spillover mechanisms could affect nearby or distant counties, but I conjecture the externalities are more likely to affect nearby projects than distant projects. A project is more likely to congest nearby transmission lines. It may trigger new grid infrastructure or congestion anywhere, but it is more likely to do so where it is located. Price effects can occur everywhere on the grid, but grid operators often use "nodal marginal prices" (i.e., local pricing), which suggest nearby counties are more susceptible to price spillovers than are distant counties.

I re-estimate all the analyses above, dropping "treated" Energy Communities, and instead comparing counties that neighbor Energy Communities as the "treatment" group to farther-away counties (Butts, 2023; Clarke, 2017).[19] Appendix F.1 reports the results.

The Energy Community Bonus has positive short-run spillover effects on nearby counties for capacity additions, especially solar and batteries. For queue entry, the results are all null, but small and positive-valued. Queued projects in counties that neighbor Energy Communities accept higher interconnection costs, but unlike in Energy Communities, this does not translate to a higher survival probability.[20] These results suggest that in the short run, the Energy Community Bonus crowded in additional investment in nearby non-Energy Communities. However, this effect does not persist in the long run. Neighbor projects have no change in queue entry or survival probability, despite the higher interconnection costs.

One takeaway from these results could be an explanation for the very large results in the main specifications in Sections 4.2 and 4.3. The inter-project externalities occur not only across borders to non-Energy Communities, but also within the subsidized areas. The new Energy Community queue entrants are not just competition; they are also reliable projects to split interconnection costs. In the case that grid infrastructure is a club good, there are conditions in

[19] A possible concern with this approach is that the neighbors have more Energy Community brownfields than the distant controls, given I observe brownfields imperfectly. For the brownfields I do observe, the neighbors and second-and-farther neighbors both average 2 brownfields per county, while the treated Energy Communities average 11 brownfields per county. So, as far as is possible to tell with the available data, this is a non-issue.

[20] This is based on initial studies only. Unfortunately, I do not have enough statistical power to measure non-first study outcomes for the neighbor group. There are not as many treatment-neighbor counties as there are Energy Communities.

which a subsidy could crowd in investment.

### 4.4.2 Distant Spillovers

The previous section shows that nearby spillovers are negligible or even positive in the short run. I argue any spillover effects of Energy Communities on more distant areas and the resultant estimate biases are likely negligible.

The spillover mechanisms driven by infrastructure—grid congestion and infrastructure costs—are especially likely to be localized. Technically, a project could trigger congestion or infrastructure upgrades anywhere, but it is much more likely to do so where it is located.

Price effects are more likely to occur everywhere on the grid. Despite the nodal marginal prices suggesting localized price effects, the grid is usually not completely congested and electricity is traded across grid operators, so nodal marginal prices are often affected by distant generation and price spillovers could have far-reaching effects. In Appendix F.2, I use a simple theoretical equilibrium model to explain why the magnitude of these far-reaching effects is trivial.

There are two reasons that price-driven spillovers are likely trivial. First, the magnitude of the spillover depends on the share of *total grid electricity generation* that is due to new capacity additions in Energy Communities. The Energy Community Bonus treatment effect is approximately two percent of total generation capacity, which is large, but not large enough to cause substantial electricity price swings. Second, the spillover effects are small if *long-run* electricity demand and supply are elastic. Most empirical evidence of electricity demand and supply elasticities are short-run elasticities, and suggest they are both inelastic. However, these results do not necessarily extend to these long-run elasticiteis. Electricity demand and supply are likely more elastic in the long run, as more margins of adjustment are available in the long run. Consistent with that intuition, Buchsbaum (2022) finds long-run residential demand elasticity of less than $-2$, and Johnson (2014) finds long-run renewable supply elasticity of over 2.

Overall, the empirical evidence suggests the net effect on nearby areas is a crowd-in in the short run and null in the long run, and distant spillovers are likely small.

## 4.5 Long-run Effects

This section combines the results on how the Energy Community Bonus affects queue entry and survival probability to calculate the long-run subsidy effect in Equation 1. To leverage the empirical estimates of the analyses in Section 4.3, I expand the survival probability of Equation 1 and write it as the joint probability of survival through the first study, all non-first studies,

and post-study survival:

$$x_j(D_1, D_2) = e_j(D_1)\rho_{0,i}(e_j(D_1), D_2)\rho_{1,i}(e_j(D_1), D_2)^{S-1}\rho_{postIA},$$

where $\rho_{0,i}(D)$ is the probability that queue entrants $i$ survive their first study, $\rho_{1,i}(D)$ is the probability queue entrants survive a non-first study, and $S$ is the number of studies a project faces as it goes through the queue, so $S-1$ is the number of non-first studies. $\rho_{postIA}$ is the probability a project does not withdraw after its interconnection agreement. I assume both $S$ and $\rho_{postIA}$ are unaffected by the subsidy, and discuss the implications of this below.

I calculate queue entry $e_j(D)$ and survival probabilities $\rho_{0,i}(D)$ and $\rho_{1,i}(D)$ using the predicted values from estimating Equations 3 and 4. I use technology-specific queue entry predictions, and aggregate technologies for survival probabilities to preserve statistical power. For confidence intervals, I bootstrap the calculation by resampling the data underlying each estimate with replacement and re-estimating the equations.

I parameterize the number of non-first studies as the mean number of studies for operating projects in the data, $S$, minus one to remove the initial study. The central estimate is 2.8, so $S-1=1.8$ non-first studies. This number may under- or over-estimate the true mean number of non-first studies that recent queue entrants will face. It is an under-estimate because projects that tend to have more studies will not yet be operating, and not be included in this mean. It is an over-estimate because the projects included in the mean are older, and their studies occurred before the recent interconnection queue process improvements. I take the mean as-is for the central estimate, but it is an uncertain parameter, so I resample it in the bootstrapping procedure.

Some projects accept their cost offers and sign an interconnection agreement, but still never become operational, usually due to permitting challenges or unexpected cost changes. I use the LBNL QueuedUp data, which has the most data available on interconnection agreement signings, to calculate $\rho_{postIA}$ from historical data (Rand et al., 2025). For projects that sign an interconnection agreement in 2017 through July 2025, I calculate $\rho_{postIA}$ as the share of operational projects among the projects that are either withdrawn or operational. I drop the half of these projects that are still active and whose post-interconnection-agreement survival or withdrawal is censored. This yields a post-interconnection-agreement survival rate of $\rho_{postIA} = 0.78$. I bootstrap this mean to account for the uncertainty this introduces.

Assuming that $\rho_{postIA}$ is unaffected by the subsidy is necessary because not enough time elapsed between the Energy Community Bonus enactment in 2023 and the 2025 repeal to directly estimate the subsidy effect on this survival rate: The median time from interconnection agreement to commercial operation for projects operating in 2025 was about 30 months (Rand et al., 2025).

This is on top of the time it takes to complete interconnection cost studies. Assuming $\rho_{postIA}$ is unaffected by the subsidy would, if anything, likely understate the magnitude of the Energy Community Bonus treatment effect, as there does not seem to be an obvious reason the subsidy would lower post-interconnection-agreement survival.

| | Solar | Wind | Battery+Other | Aggregate |
|---|---|---|---|---|
| MW/county/year | 13.2 | 3.3 | 14.5 | 31.6 |
| | (2.2, 26.8) | (-0.8, 9.2) | (3.8, 30.5) | (7.2, 65.2) |
| Percent increase | 259 | 135 | 364 | 289 |
| | (43, 527) | (-33, 373) | (96, 765) | (66, 596) |

Table 4: Long-run effect of the Energy Community Bonus on annual capacity additions from Equation 1. Percent increase is relative to the predicted counterfactual without the subsidy. Bootstrapped 95 percent confidence intervals in parentheses.

Table 4 shows the result. In the long run, the Energy Community Bonus increases capacity additions in Energy Communities by about 30 MW per county per year. The percent increases are expressed as a percent of the counterfactual $(x_j(1,1) - x_j(0,0))/x_j(0,0)$. They show nearly a 289 percent increase in capacity additions in Energy Communities due to the Bonus in the long run. Most of this is driven by solar and battery storage. The Bonus increases aggregate queue entry from 86 MW per county per year to 164 MW per county per year, a 92 percent increase. It increases the predicted survival probability from 13 percent to 26 percent.

The combined effect is 32 MW per county per year, or a 289 percent increase from the baseline. The implied aggregate treatment effect is between 23 GW and 47 GW per year. The lower bound uses a strict definition of Energy Communities, and the upper bound a more inclusive definition.[21] Given battery and solar almost exclusively take the ITC, and the Energy Community Bonus increases the ITC from 30 to 40 percent, this implies a subsidy elasticity of about 8 for solar and 11 for batteries. Wind typically takes the PTC, which increases by 10 percent, so the implied subsidy elasticity is about 13.

These long-run estimates capture how the queue entrants in the immediate years after the Energy Community Bonus enactment would have converted to capacity additions, were the subsidy not repealed for solar and wind. It is still possible that this does not reflect a permanent annual

[21] Specifically, the lower bound counts only counties that are fully covered by coal mine or generator closure census tracts or qualify by the fossil employment designation in all years 2023 to 2025. The more inclusive upper bound adds counties that contain at least one coal census tract or have more than ten brownfields.

effect of a subsidy if it were left in place indefinitely. Perhaps there are dynamic incentives such that the subsidy causes a short-run spike in Energy Community queue entry or survival that would not persist. Furthermore, electricity demand and supply are regularly changing due to macroeconomic factors and technology. Even in light of these concerns, this long-run effect still captures how the Energy Community Bonus would have affected capacity for the foreseeable future.

The remainder of the paper combines these empirical results with data on electricity emissions to measure the emissions and welfare effects of the Energy Community policy and clean electricity subsidies generally.

# 5 Emissions and Welfare Effects

I use two approaches to measure the welfare effects of clean electricity subsidies. The first approach uses the electricity quantity changes from the previous analyses as sufficient statistics for welfare changes, based on the simple intuition of Harberger triangles (Chetty, 2009). The second approach uses the MVPF (Hendren and Sprung-Keyser, 2020; Hahn et al., 2026).

I use the empirics and models to estimate the marginal welfare effects of clean electricity subsidies in the US. I then assess how the Energy Community Bonus compares to other policies in terms of emissions abated, welfare effects, and the MVPF. But first, I offer some general intuition about the welfare effects of place-based clean electricity subsidies.

## 5.1 The Welfare Effects of Place-Based Clean Electricity Subsidies

In the case of a carbon tax or a subsidy per carbon abated, it is clear that a uniform policy is more cost-effective than a place-based policy. Indeed, the whole point of carbon pricing is to equalize abatement costs, and spatially differentiating the instrument undermines this. In the case of clean electricity subsidies, it is not obvious whether the place-based subsidy will improve or worsen the welfare relative to some counterfactual subsidy (e.g., a capacity-neutral or fiscally neutral subsidy). The reason is that clean electricity is an imperfect proxy for emissions abatement. The ITC is a subsidy per MW and the PTC a subsidy per MWh, but the externality per MW or MWh spatially varies throughout the country. Therefore, an abatement-cost minimizing subsidy also spatially varies.

There are two sources of spatial variation that drive the imperfection. The first is spatial variation in marginal emissions. Marginal emissions are the emissions that are generated by the marginal generator on the electricity grid (usually the most expensive generator), and that are

displaced from an additional unit of lower-marginal-cost electricity, such as renewables. Clearly, generators do not internalize the social value of emissions abatement, so pricing proportionately to marginal emissions can improve outcomes.

Perhaps less clearly, the other source of imperfect pricing is geography, in particular windiness and sunniness. This is perhaps less clear because generators do, at least to some extent, internalize this value. Consider second-best optimal clean electricity subsidies in a case where marginal emissions are spatially uniform.[22] The second-best optimal ITC spatially varies proportionately to sunniness and windiness, so a uniform instrument is surely suboptimal. But the second-best optimal PTC is uniform. Nonetheless, a marginal increase in ITC or PTC levels in sunnier or windier areas still improves welfare compared to a marginal increase in a less sunny or less windy area. The reason is that the baseline PTC is below the second-best optimal level, at least according to my calculations of subsidy values and externality values below. So a marginal increase to the PTC in a windy area improves welfare more than a marginal increase in a non-windy area. Appendix G.1 shows this using a simple model.

With this intuition, whether the Energy Community Bonus improves or worsens welfare relative to counterfactual subsidies depends on the correlation between Energy Community designation, marginal emissions, and geography. I next describe a framework to quantify the welfare effects of clean electricity subsidies using the empirical results in this paper.

## 5.2 Model

### 5.2.1 Sufficient Statistics for Welfare

Denote $x_j$ as a county-technology combination of a clean electricity capacity addition. For example, one unit of $x_j$ may be one MW of solar capacity in some county. Let $\phi_j$ be the net present value (NPV) of an externality caused by one unit of $x_j$.

Consider a marginal change moving from one tax vector $\tau = (t_1, ..., t_J)$ to another tax vector $\tau' = (t'_1, ..., t'_J)$. The change in deadweight loss is

$$\Delta DWL = \frac{1}{2} \sum_j \sum_k (\phi_j - t_j + \phi_j - t'_j) \frac{dx_j}{dt_k} (t'_k - t_k). \tag{5}$$

The formal proof is in Appendix G.3. The intuition is from the Harberger triangle (Harberger, 1964). This formula is for the trapezoid that is formed as the difference between two deadweight loss (DWL) triangles. $\phi_j - t_j$ and $\phi_j - t'_j$ are the heights of the two magnitudes of mispricing

[22] "Second-best optimal" here refers to a policy that can only subsidize clean electricity in an economy where there is no other direct or indirect carbon pricing.

caused by $t_j$ and $t'_j$. $\frac{dx_j}{dt_k}(t'_k - t_k)$ is the quantity change in $x_j$ caused by a change to the tax on $x_k$, the width of the trapezoid. The sum is over every county-technology combination, where the terms for $j \neq k$ are spillover effects. Similarly, the change in emissions caused by the policy change is

$$\Delta Emissions = \sum_j \sum_k e_j \frac{dx_j}{dt_k}(t'_k - t_k)$$

where $e_j$ is measured in NPV since the change in emissions caused by the new clean electricity capacity is a stream of emissions changes into the future.

Importantly, the quantity change partial $\frac{dx_j}{dt_k}$ is not a demand or supply elasticity per se. It is the quantity change that results due to the net of supply and demand. This is the key benefit of writing the expression for the change in DWL this way. The quantity changes from the empirical analysis are sufficient statistics for welfare effects.

Deriving Equation 5 makes assumptions which have implications for the types of counterfactuals I can model. The first assumption is that the externalities $\phi_j$ are constant over the support of the change in $x_j$ for all $j$. Thus, the model assumes the marginal generator is unchanged by the presence or absence of the Energy Community Bonus or any of the counterfactual policies.

The second assumption is that $\frac{\partial x_j}{\partial t_k}$ are constant. A consequence of this linearity assumption is that it implies that an increase in the subsidy rate linearly increases clean electricity, while convexly increasing fiscal costs.[23]

This undermines using a uniform, revenue-neutral subsidy counterfactual when evaluating the place-based subsidy. The revenue-neutral subsidy will dominate almost any place-based policy, barring extreme externality differences. Spreading the fiscal cost over more locations will cause more subsidy effect ($\frac{dx_j}{dt_j}$) per fiscal dollar because there is less fiscal cost penalty from the convexly increasing fiscal costs in any particular $t_j$. That means more renewable electricity and externality per dollar under any subsidy that covers more locations.

For a counterfactual comparison that is not driven by the linearity assumption, instead of comparing the Energy Community Bonus to a revenue-neutral subsidy, my preferred counterfactual benchmark is a capacity-neutral subsidy. This is a uniform subsidy that results in an equal amount of capacity as the Energy Community Bonus. The capacity-neutral subsidy is calcu-

[23] Fiscal costs are expressed as $\sum_j \sum_k t_j \frac{dx_j}{dt_k} t_k$. Assuming no spillovers, it is easy to see the convex functional form $\sum_j \frac{dx_j}{dt_k} t_j^2$. In terms of the Harberger triangle, this is the revenue rectangle increasing in both height and width with a subsidy increase.

lated as

$$\tilde{t} = \frac{\sum_j \frac{dx_j}{dt_j} EC_j t_j^{EC}}{\sum_j \frac{dx_j}{dt_j}},$$

where $t_j^{EC}$ is the Energy Community bonus value for $j$ (not including the baseline ITC or PTC values) and $EC_j$ is equal to 1 if county $j$ is an Energy Community. Note, this is not only the capacity-neutral uniform subsidy, but also the quantity-change-weighted mean value of the Energy Community Bonus across all counties and technologies. I will show below using data that the subsidy value depends on technology types. I therefore calculate these values, and use as counterfactuals the within-technology capacity-neutral subsidies.

### 5.2.2 Marginal Effects Model

The MVPF is a benefit-cost ratio,

$$MVPF = \frac{WTP}{Net\ Govt\ Cost} = \frac{x \cdot d\tau + \phi \cdot dx}{x \cdot d\tau + \tau \cdot dx} \tag{6}$$

where the numerator is societal willingness to pay (WTP) for a policy and the denominator is the net cost to the government. $x$, $d\tau$, $\phi$, $dx$, and $\tau$ are $J$-length vectors, and the dot is the dot product. The components are the same as those in Equation 5, though this adds the requirement of knowing the size of the inframarginal capacity additions $x_j$.

Societal WTP for a marginal transfer is the magnitude of the transfer $x \cdot d\tau$ to inframarginal recipients, plus the externality created by the transfer $\phi \cdot dx$, where $dx$ is the behavioral change caused by the policy. The marginal transfer recipients do not value the transfer because of the market distortion induced; they are exactly indifferent between receiving the transfer and the marginal unit of consumption (the envelope theorem). The fiscal cost in the denominator is composed of the same marginal transfer to the inframarginal recipients, plus the total value of the transfer to the marginal recipients.

The MVPF has the desirable property that a budget constrained social planner choosing between two policies would raise the most welfare by choosing the policy with the highest (welfare-weighted) MVPF. A policy with an MVPF greater than one raises social welfare, and a policy with an MVPF less than one lowers welfare. The MVPF is proportionate to the wedge between social marginal benefit (cost) and prevailing marginal benefit (cost). A marginal subsidy at the efficient quantity, or a Pigouvian subsidy, has an MVPF equal to one. A marginal subsidy that exceeds the efficient quantity has an MVPF less than one, and a marginal subsidy below efficient quantities has an MVPF greater than one.

Below, I consider an MVPF for raising the subsidy level by \$1 for all counties and all technologies,

$$MVPF = \frac{\sum_j x_j + \phi_j \sum_k \frac{\partial x_j}{\partial t_k}}{\sum_j x_j + t_j \sum_k \frac{\partial x_j}{\partial t_k}}. \tag{7}$$

I also consider another metric of interest to benchmark clean electricity subsidies against other climate policies, the marginal fiscal cost per ton of $CO_2$ abated from raising the subsidy level by one dollar:

$$\frac{\$Subsidy}{Tons} = \frac{\sum_j x_j + t_j \sum_k \frac{\partial x_j}{\partial t_k}}{\sum_j \sum_k \frac{\partial x_j}{\partial t_k} e_j}, \tag{8}$$

where $e_j$ is again the NPV of the emissions abated by an additional MW of $x_j$.

Like the welfare calculation in Equation 5, calculating the MVPF in Equation 7 does not require an estimate of supply or demand elasticities. All that is needed is externality values, subsidy values, and estimates of the marginal effect of a subsidy change on capacity additions and the inframarginal capacity additions. The next section describes how I measure each of these components.

## 5.3 Data and Calculations for Emissions and Tax Credit Values

As alluded to in Section 3, I use forecast data of future electricity grid emissions, electricity prices, and capacity factors to calculate clean electricity externalities and tax credit values. I describe the process and calculation results in this section, with more detail available in Appendix G.2.

**Emissions Abated by Clean Electricity ($\phi$)** Calculating the NPV of the emissions abated by one MW of technology-location-specific electricity capacity has three key ingredients: (1) the future stream of electricity generated by the MW of capacity (converting MW to MWh), (2) the emissions abated by that MWh (converting MWh to emissions), and (3) the value of those emissions. I calculate this stream of electricity and emissions at the county-by-hour level for every county in the US and every hour from 2025 to 2050.

For electricity generated by a MW, or capacity factors, I use 15 years of historical sunniness and windiness data to obtain the electricity generated by the optimal generator choice for each location based on data from the National Renewable Energy Laboratory (NREL) ReEDS model (Sergi et al., 2024). I use the Cambium 2024 model for a future stream of electricity prices, assuming batteries charge during the least expensive four hours and discharge during the most expensive four hours per day (Gagnon et al., 2024). To convert the generation to emissions abated, I use the Cambium model's marginal emissions projections.

For emissions results, I convert the stream of future emissions abatement to a present discounted value (PDV) assuming a 2 percent social discount rate on emissions abated. For results that value the emissions externality, I first multiply the Social Cost of Carbon (SCC) from EPA (2023) by the stream of abatement, and then discount the stream of valued abatement to a PDV.

**Tax Credit Values ($\tau$)** I calculate the value of the baseline ITC and PTC and the Energy Community Bonus adder for one MW of solar, wind, or batteries in each county. Projects can choose which of the tax credits to claim. I calculate both tax credit values and assume the project takes whichever is most valuable per MW.

To calculate the ITC value, I use the 2024 Annual Technology Baseline model, which has the capital expenditures required per MW of solar, wind, or battery storage (NLR, 2024). The model chooses the solar panel and wind turbine that it estimates is most profitable at each location. The ITC baseline is 30 percent of this value, and the Energy Community Bonus an additional 10 percent.

Calculating the PTC value per MW is more complicated because it requires converting a 10-year subsidy per MWh to a subsidy per MW in PDV. I assume that project developers use a private discount rate of 8 percent when valuing the 10-year stream of values. For the purpose of weighing the PTC values against the externality values, I use a social discount rate of 2 percent. I use the same capacity factors as I do for marginal emissions to obtain 10 years of electricity generation per MW for every technology and every location. The PTC is a constant 2.75 cents per kWh in 2023, indexed to inflation, which I assume to be 2.5 percent per year based on the 2025 30-year inflation expectations (FRED, 2025).

I use a more inclusive definition of Energy Communities for these welfare and emissions analyses than I do for the empirical analyses. The empirical analysis uses the strict definition of Energy Communities for an unbiased county-level subsidy effect estimate. However, even partial Energy Communities may see some subsidy effect. Here I consider Energy Communities to be counties that meet the fossil fuel employment definition in all years, have at least one coal census tract, or at least 10 brownfields. Under this definition, 1,499 counties are Energy Communities, which is 48 percent of the counties in the US.

Appendix Figure G.1 shows the result of the calculations. Wind takes the PTC and battery the ITC with rare exception. Solar takes the PTC in the sunniest parts of the country, and otherwise takes the ITC. The Energy Community Bonus is a larger adder to the ITC (10 percentage points) than the PTC (10 percent), so it makes the ITC the more valuable credit regardless of location

for solar. It also nudges wind to claim the ITC in a few locations. The Energy Community Bonus value ranges from $57 thousand to $215 thousand. It is most valuable for locations and technologies that claim the ITC. The average private PDV of the Bonus per MW for batteries is $171 thousand, for solar is $146 thousand, and for wind is $77 thousand.

**Inframarginal Capacity Additions $x_j$ and Marginal Policy Effects ($\frac{\partial x_j}{\partial t_k}$)** Based on the empirics and argument in Section 4.4, I assume zero spillover effects such that $\frac{\partial x_j}{\partial t_k} = 0$ for all $j \neq k$. This greatly simplifies the welfare Equation 5 and the MVPF Equation 7. I use the aggregate long-run treatment effects (i.e., from Table 4) as the national average treatment effect.

I incorporate location-by-technology heterogeneity for the average aggregate estimate based on the historical interconnection queue data in each area. The resulting average share of each technology is 43 percent solar, 17 percent wind, and 40 percent battery with sensible regional variation. The long-run aggregate treatment effect in Table 4 is about 42 percent solar, 11 percent wind, and 46 percent battery and other. The consistency of these values supports the approach assuming a uniform treatment effect varying according to technology prevalence.

I calculate $\partial x_j / \partial t_j = \hat{\beta}_j / t_j^{EC}$, where $\hat{\beta}_j$ is the aggregate treatment effect multiplied by county $j$'s share of that technology and $t_j^{EC}$ is the county-technology-specific Energy Community Bonus value not including the baseline tax credit. The inframarginal capacities $x_j$ are the predicted counterfactual capacity additions.

## 5.4 Results

Table 5 presents the national MVPF estimates (Equation 7) and marginal fiscal cost per ton (Equation 8). The MVPF is highest for solar and wind, and lower for batteries. This difference is driven by the fact that batteries need to charge, so they both emit and abate. However, the MVPF for all technologies is quite high. For example, over $7 of social benefit is created for every $1 of fiscal transfer toward a wind subsidy. This MVPF is higher than the equivalent MVPF in Hahn et al. (2026), about 3.6, so using an updated policy elasticity appreciably raises the MVPF.[24] Marginal fiscal costs per ton of $CO_2$ provide the same policy ordering as the MVPF. By raising all the subsidy levels by $1, the resultant fiscal transfer per ton of $CO_2$ is $38.

### 5.4.1 A Trade-off or Win-Win for Efficiency and Emissions?

Table 6 presents the results calculating Harberger triangles and emissions abated by the Energy Community Bonus compared to a nationwide uniform subsidy that results in an equal amount

[24]The 3.6 MVPF is based on their Figure 2.A, counting only the global environmental externality value (4.931/1.354).

| | Solar | Wind | Battery+Other | Aggregate |
|---|---|---|---|---|
| MVPF | 8.56 | 7.36 | 2.60 | 6.49 |
| | (6.11, 9.11) | (6.62, 7.48) | (2.06, 2.73) | (5.12, 6.77) |
| Marginal subsidy per ton ($) | 29.07 | 33.24 | 105.56 | 38.37 |
| | (27.07, 42.40) | (32.60, 37.49) | (97.73, 157.54) | (36.49, 50.84) |

Table 5: MVPFs and marginal fiscal cost per ton as described in Equations 7 and 8 and 95 percent confidence intervals. Confidence intervals generated using the bootstrapped estimates of the policy effect from the aggregate effect estimates.

| | ECB − Capacity Neutral Uniform | | |
|---|---|---|---|
| | Solar | Wind | Battery |
| Emissions abated (Mt $CO_2$e) | 5.1 | 5.8 | -3.1 |
| Welfare from abatement ($B) | 0.1 | 0.7 | -1.1 |

Table 6: Welfare effects of the Energy Community Bonus compared to a capacity-neutral uniform subsidy. Values are in NPV of emissions or welfare caused by one year of long-run capacity effect.

of new capacity of solar, wind, and battery technology. The Energy Community Bonus improves the emissions and welfare effects of solar and wind subsidies. It worsens the effects for battery subsidies. For every year the Energy Community Bonus is in place for solar and wind, it results in an additional $800 million of NPV of annual social surplus. However, the equivalent loss is $1.1 billion per year for batteries. So, the net loss is $300 million per year that the Energy Community Bonus is enacted. With the repeal of solar and wind subsidies, the loss is $1.1 billion per year. The magnitude of the battery welfare loss is higher than the relative magnitude of its emissions. This is because the welfare calculation weights the effect not only by the SCC, but also by the deviation from the social cost, including the baseline tax credit. The Energy Community Bonus causes a larger shift away from the social cost for batteries than it does for solar and wind combined.

Figure 8 assesses the question in two other ways, with similar qualitative results. First, I plot the national MVPF of a $1 subsidy increase, and the MVPF of an Energy-Community-only $1 subsidy increase. This compares the MVPF of a marginal place-based subsidy to a marginal national subsidy. To get a sense of magnitude, the histogram shows the MVPF of a $1 subsidy increase to 1,499 random counties (the same number as there are Energy Communities), iterated 10,000 times. The Energy Community Bonus is better than random for solar, wind,

and the aggregate subsidy. For batteries, it is within the tails of the random histogram, but still substantially worse than random.

The intuition for the Bonus improving solar and wind effects and worsening battery effects is that, on average, the marginal electricity in Energy Communities tends to be more emissions-intensive than the marginal electricity elsewhere in the US. So, solar or wind electricity offsets more emissions-intensive electricity and has a higher social gain. For batteries, although they abate more emissions-intensive electricity when discharging, they also charge using more emissions-intensive electricity. This could be net beneficial or costly, but it turns out to be net costly.

These results capture subsidy welfare effects through their effects on greenhouse gas emissions. Future work could also incorporate welfare effects through local air pollution, local renewable generator disamenities, and learning by doing. It could also capture the possibility that subsidizing new renewable generators could lower equilibrium electricity prices and cause more electricity consumption, a so-called rebound effect. However, previous work on electricity subsidies suggests welfare is more affected by $CO_2$ abatement than by air pollution, learning by doing, or rebound effects (Hahn et al., 2026). Furthermore, these welfare effects do not account for other state-level policies such as California's cap and trade and the Regional Greenhouse Gas Initiative. Future work could account for how these policies interact with subsidies, and potentially change the welfare results.

# 6 Conclusion

This paper shows that increasing utility-scale clean electricity subsidies causes large increases in new electricity capacity. This counters the narrative that permitting delays, local opposition, or institutional barriers preclude subsidy effects. On the contrary, a subsidy increase leads to a nearly immediate increase in renewable generating capacity. These short-run increases may not be subject to some points underlying the skepticism, such as interconnection queue bottlenecks and congestion. But I show that subsidies increase both interconnection queue entry and survival, so the skepticism that institutional or infrastructure barriers would limit the long-run subsidy effectiveness also seems unfounded.

Furthermore, increasing clean electricity subsidies causes large increases in welfare. This is because the social value of the emissions abated by clean electricity is much greater than the prevailing subsidy levels. The Energy Community Bonus raises the welfare effects of solar and wind subsidies, but reduces the welfare benefits of battery subsidies. However, zooming out, even the worst targeted clean electricity subsidies still increase societal welfare and have high MVPFs, again because of the large difference between the value of emissions and prevailing

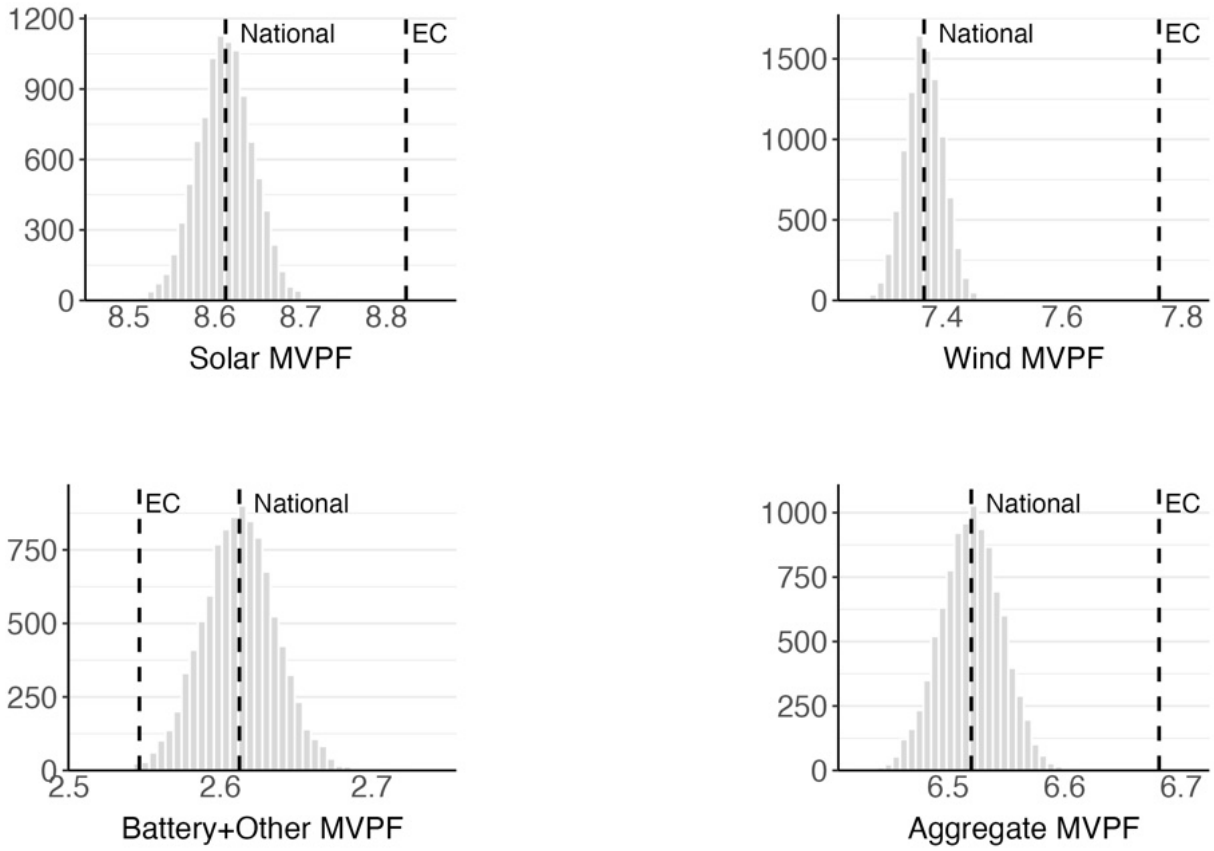



Figure 8: The MVPF of an additional dollar to Energy Communities versus a random choice of counties.

subsidy levels.

The high welfare value of clean electricity subsidies relies on carbon emissions being unpriced elsewhere in the US economy. Given that carbon emissions are not priced in the US, and extending the lessons from this marginal increase in subsidy, the recent repeal of solar and wind subsidies could substantially reduce solar and wind capacity, increase emissions, and reduce welfare.

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# A Interconnection Queue Process

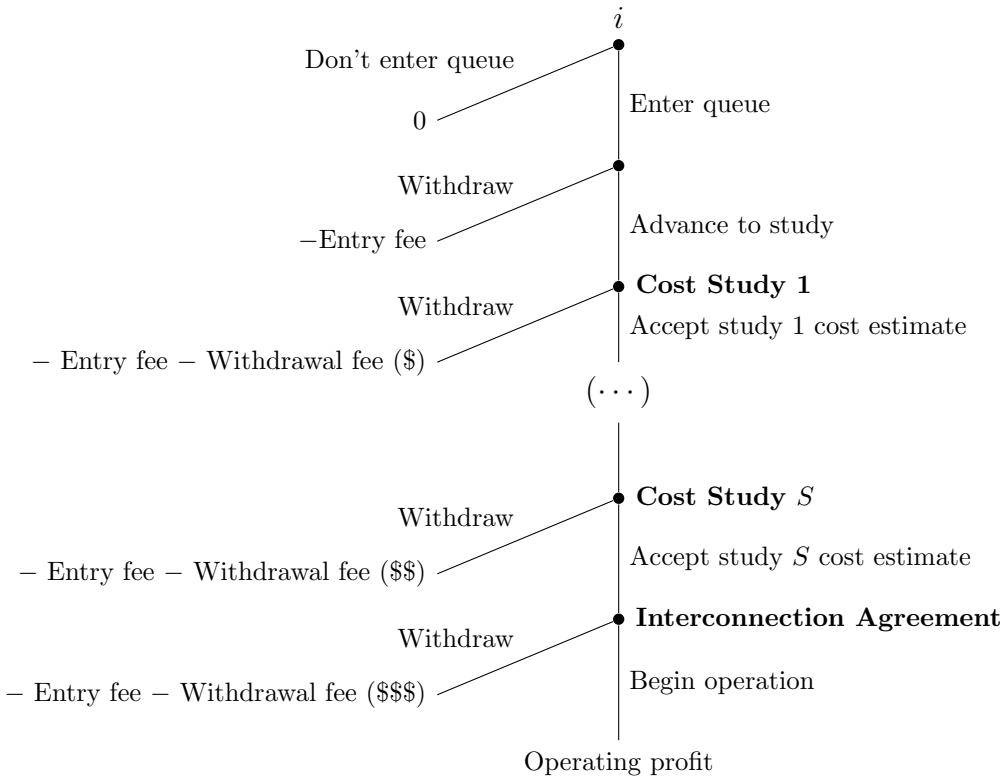



Figure A.1: Interconnection queue process as a game tree.

# B Data Cleaning

## B.1 Interconnection Queue Data Cleaning and Missing Data Handling

The queue data include observations from all contiguous 48 states. I drop observations in Mexico and Canada from the estimating sample. I also filter out load and transmission projects that are in queues because these are not eligible for the ECB. That leaves 37,535 observations of projects from 1995 to August 2025. I need all queued projects to have location data precise enough to join them to county-level Energy Community data. Most projects list FIPS codes that are sufficient for this join. 2 projects have no FIPS codes, but include coordinate data that I use to join the projects to counties. 12 projects are associated with cities, not counties, so I manually join them to their containing county. 563 observations have no data on FIPS codes, coordinates, county names, or city names. I drop these observations, and am left with 36,972 projects.

The data lists multiple FIPS codes for 336 of projects. However, it is important to connect projects to one county, as only one county can determine its EC status.[25] This adds an additional dimension of uncertainty regarding whether these projects are indeed EC projects. For project-level analyses, I split multi-county projects into multiple observations, and weight these observations by the inverse of the number of counties associated with the project. For example, a project that lists two counties is associated with both counties, but only included in regressions as a half-project. For county-level analyses, I again split multi-county projects into multiple observations. I count fractional projects for to each county, and evenly divide the MW of the multi-county project across the listed counties.

Another issue that needs to be addressed with the Interconnection.FYI dataset is missing data. Interconnection.FYI collects data by scraping balancing authority websites and, when necessary, directly contacting balancing authorities. The result is coverage of grid operators that account for about 97 percent of electricity capacity. However, that leaves about 3 percent of US electricity generation missing from the data (Rand et al., 2025). For the county-level queue entry analysis, I identify and drop counties that are covered by the grid operators that are not sampled by the Interconnection.FYI data. Without this correction, the count of projects in MW in these counties would be biased downward. I use data from the US Department of Energy (DOE) with the approximate spatial coverage area of balancing authorities in the US to identify and drop

[25] A project is considered an EC project if 50 percent or more of its nameplate capacity or its footprint is in an EC. For offshore generation, this is determined by where the equipment connects to land (Service, 2025; Treasury, 2023a; Treasury, 2023b).

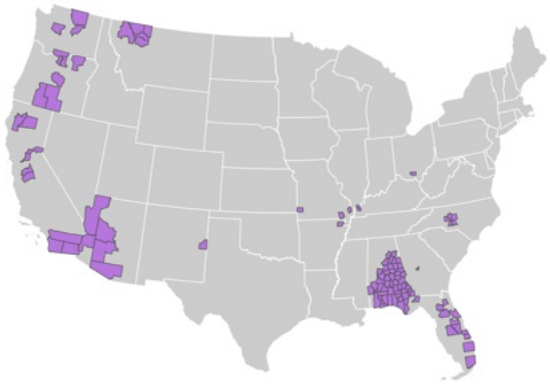


Figure B.1: Counties covered by balancing authorities that are not in the data sample.

counties that are covered by unsampled balancing authorities.[26] Specifically, I find balancing authorities in the DOE data that are not in the interconnection queue data or listed in the Interconnection.FYI metadata on grid operators covered. This drop 107 counties, shown in Figure B.1.

A second type of missing information in the Interconnection.FYI is somewhat more consequential. All of the sampled balancing authorities include information on project status (whether it is active, withdrawn, or operational), but some do not include the dates of status changes. These dates are crucial for the analyses in this paper, as identification relies on the timing of ECB policy enactment and the timing of status changes. For queue entry analyses, I drop counties covered by ERCOT, Georgia Transmission, Dominion Energy South Carolina, or Idaho Power Company. ERCOT in particular contains almost 20 percent of the active capacity in US queues as year-end 2025. Although the ERCOT website updates is queue data monthly, this is apparently not a reliable way to track new queue entrants. For example, some projects are recorded as signinig an interconnection agreement in August 2024, but do not appear in ERCOT's queue reports until August, September, or October 2024.

For dropping counties, I assess whether a county is managed by a certain grid operator in two ways, and drop the counties if it is identified by either method. First, I consider counties with a queued project managed a balancing authority as covered by that balancing authority. Second, I use the DOE spatial data and select counties that have more than 1 percent of their area covered by a balancing authority with a lot of missing data. The procedure drops ultimately drops 586 counties for the queue entry analyses. After this procedure, 0.6 percent of the remaining projects

[26] https://atlas.eia.gov/datasets/balancing-authorities/explore

have a null queue entry date. These are dropped from the county-level queue entry analyses. The final estimating sample for county-level queue entry analyses includes 28,239 projects and 2,522 counties.

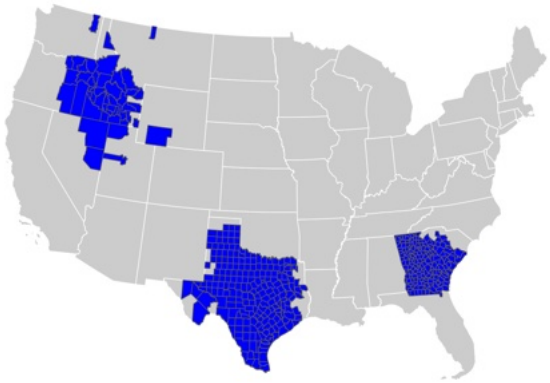


Figure B.2: High proportion null queue entry dates.

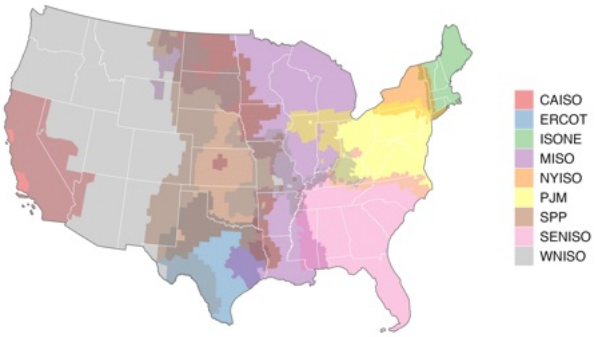



Figure B.3: Counties managed by each grid operator.

## B.2 Energy Community Data Cleaning

Treasury (2023a), Treasury (2023c), Treasury (2023d), Treasury (2023e), and Treasury (2023b) define the Energy Community definition data. The list of NAICS codes used to define fossil fuel employment as it pertains to Energy Communities was expanded in March 2024, over a year after the Bonus had been made available. I use the initial fossil fuel employment designation for defining Energy Communities in 2023 and earlier, and the updated definition in 2024 and later.

Related, it is somewhat ambiguous which is the correct County Business Patterns (CBP) data to use. CBP for a given year is usually released with about a two year lag. In 2023, it is clear

that the official EC definitions used the 2021 CBP, as that was the most recent CBP available. When the 2024 official EC dataset was released, the 2021 CBP was still the most recent version of the CBP, so it must also consider the 2021 CBP. The 2022 CBP was released a few months later. However, the 2022 CBP would have been available for tax year 2024. I use the most recent CBP data that was available in the tax-year. It turns out this does not matter because no MSA or NMSAs newly pass the 0.17 percent fossil fuel employment threshold in 2024 or 2025. But it is perhaps notable for future work that attempts to identify Energy Communities.

To identify EC Brownfields in the EPA Facility Registry service, I follow the Treasury Guidances, as well as the EnergyCommunities.gov Frequently Asked Questions page from 2024, which included a question comparing EPA brownfields to Energy Community Brownfields. I drop observations that are Superfund sites and observations associated with the Resource and Conservation Recovery Act, both of which are not eligible Energy Community brownfields. Parcels that only have pollution due to petroleum contamination are also ineligible, so I drop observations that are likely a gas station or petroleum processing facility. I drop facilities with primary names that include the name of about a dozen of the largest gas station brands in the US, or include the words petroleum, gasoline, or gas station (e.g., Mobil, BP, etc.).

Even if this process perfectly identifies the subset of EPA Brownfields that are also EC Brownfields, it will still miss locations that may be eligible as Energy Community Brownfields and have not yet been identified by an IRS environmental assessment. However, given that Brownfields tend to be spatially clustered, this measure should give a good proxy for Brownfields even if it is imperfect.

# C Treatment Heterogeneity by Grid Operator

I re-run the SDID at the grid operator level, accordingly dropping the operator-by-year fixed effects. Many ISOs have an insufficient number of treatments or controls, so I group them based on geographic proximity. Unsurprisingly, running estimating treatment effects for these sparse grid operators alone produces noisy and null results. I group PJM, ISO-NE, and NYISO because ISO-NE and NYISO combined only have 8 treated Energy Community counties, 4 neighboring counties to investigate spatial spillovers, and about 32 controls. CAISO only has 3 eligible control counties, and ERCOT has 80 Energy Community counties compared to 20 controls, so I group them both with non-ISOs. Jackknife standard errors are in parentheses, following Arkhangelsky et al. (2021) Algorithm 3. Bootstrapping is no longer necessary because there is not first stage fixed effect estimation.

Results are in Table 7 for Energy Community direct effects, and Table 8 for neighboring county spillover effects. Each cell in this table is a separate SDID run. For legibility, I include asterisks for significance according to multiple hypothesis testing Bonferroni-adjusted p-values. I treat these as 32 tests, one for each cell within the tables.

For effects on Energy Communities, the positive US-level treatment effects seem to be driven by MISO, SPP, CAISO, ERCOT, and non-ISOs. There are no significant effects in the northeast ISO regions.

Neighbor spillover effects are noisier due to a smaller number of treated counties, especially SPP. Nothing is significant. A couple positive queue entry spillovers have their significance undermined by the multiple hypothesis testing correction, including solar queue entry in PJM, NYISO, and ISO-NE, and battery queue entry in MISO, and aggregate technology capacity additions in MISO, CAISO, ERCOT, and non-ISOs.

Overall, these results suggest heterogeneity, but also a lot of insignificance and limited statistical power. In assessing subsidy effects, I advocate for using the full US level sample because the most suitable control counties are not necessarily in the same ISO, and pooling the regions offers more statistical power.

| | | MISO | SPP | PJM, NYISO, ISO-NE | CAISO, ERCOT, non-ISO |
|---|---|---|---|---|---|
| Cap. Add. | Solar | 10.8*** | 17.6 | 1.0 | 16.9*** |
| | | (2.6) | (8.2) | (2.1) | (3.9) |
| | Wind | 1.9 | -1.4 | -0.6 | -0.7 |
| | | (1.3) | (3.5) | (0.6) | (1.9) |
| | Battery | 2.3 | 4.1 | -0.2 | 13.5** |
| | | (1.3) | (2.8) | (0.3) | (3.8) |
| | Aggregate | 14.1*** | 21.2 | -0.6 | 31.5*** |
| | | (3.5) | (10.9) | (2.4) | (7.1) |
| | N treated / control | 327 / 353 | 85 / 401 | 244 / 166 | 360 / 597 |
| Queue | Solar | 24.8*** | 23.1 | -1.6 | -1.2 |
| | | (5.2) | (14.7) | (4.1) | (10.4) |
| | Wind | -5.5 | 15.3 | -0.6 | -13.1 |
| | | (6.5) | (12.3) | (2.2) | (11.4) |
| | Battery | 13.5*** | 23.0 | -13.7 | 16.7 |
| | | (2.6) | (11.6) | (7.0) | (11.4) |
| | Aggregate | 19.1 | 56.6 | -14.9 | 36.2 |
| | | (9.1) | (30.0) | (10.4) | (23.3) |
| | N treated / control | 292 / 351 | 49 / 388 | 243 / 166 | 242 / 386 |

Table 7: Energy Community treatment effect results, disaggregating grid operators. Capacity addition outcomes are in MW per county per year. Queue entry outcomes are in MW per county per half-year. Jackknife SEs in parentheses. Discrepancies between queue entry and operating regressions are also due to missing queue data or queue date data. Most notably, ERCOT drops out of the queue entry results.

| | | MISO | SPP | PJM, NYISO, ISO-NE | CAISO, ERCOT, non-ISO |
|---|---|---|---|---|---|
| Cap. Add. | Solar | 7.2 | 4.6 | 7.0 | 3.6 |
| | | (3.7) | (4.1) | (6.7) | (3.6) |
| | Wind | 2.9 | -3.6 | -2.9 | 0.7 |
| | | (2.6) | (4.5) | (2.3) | (2.4) |
| | Battery | -0.2 | -0.5 | -0.1 | 4.4 |
| | | (0.6) | (0.4) | (0.6) | (2.2) |
| | Aggregate | 9.5 | 0.8 | 5.0 | 10.6 |
| | | (4.9) | (5.7) | (9.1) | (5.6) |
| | N treated / control | 87 / 353 | 27 / 401 | 45 / 166 | 120 / 597 |
| Queue | Solar | 2.8 | -8.0 | 3.5 | 0.3 |
| | | (8.2) | (3.8) | (5.8) | (6.0) |
| | Wind | -2.0 | -10.3 | -9.7 | -0.3 |
| | | (5.6) | (14.3) | (9.1) | (4.0) |
| | Battery | 6.7 | -1.0 | -5.7 | 0.0 |
| | | (3.2) | (5.3) | (8.1) | (4.9) |
| | Aggregate | 2.4 | -30.0 | -18.0 | -2.7 |
| | | (10.1) | (21.5) | (17.4) | (14.0) |
| | N treated / control | 86 / 351 | 24 / 388 | 45 / 166 | 94 / 386 |

Table 8: Neighbor spillover effect results, disaggregating grid operators. Capacity addition outcomes are in MW per county per year. Queue entry outcomes are in MW per county per half-year. Jackknife SEs in parentheses. Discrepancies between queue entry and operating regressions are also due to missing queue data or queue date data. Most notably, ERCOT drops out of the queue entry results.

# D Robustness Checks for Queue Entry and Capacity Additions

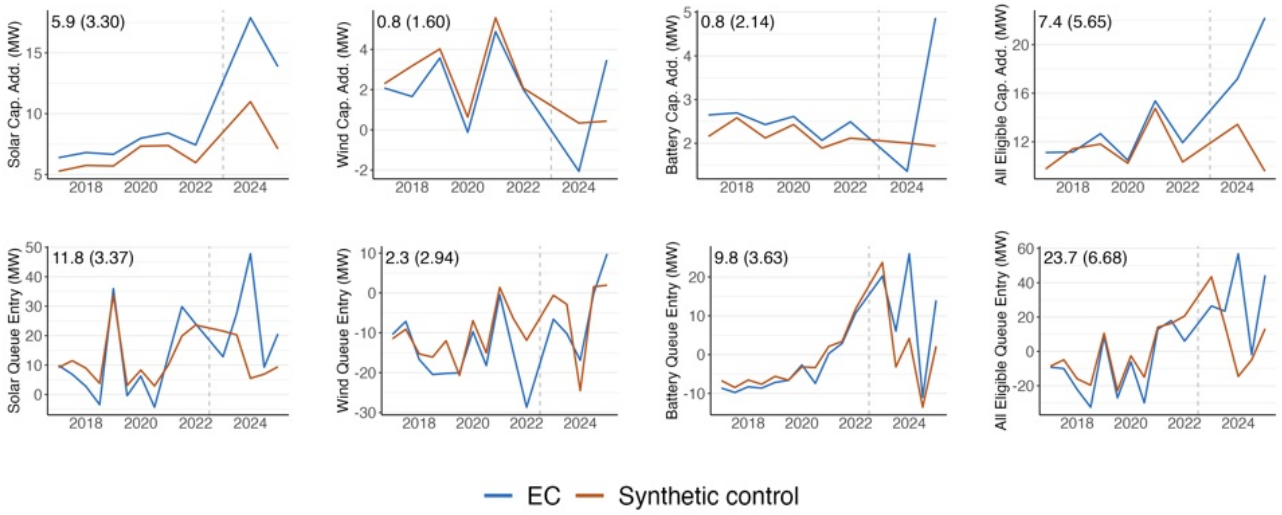



Figure D.1: Capacity addition (top row) and queue entry (bottom row) estimating SDID Equation 3, dropping Arizona, California, Nevada, and New Mexico.

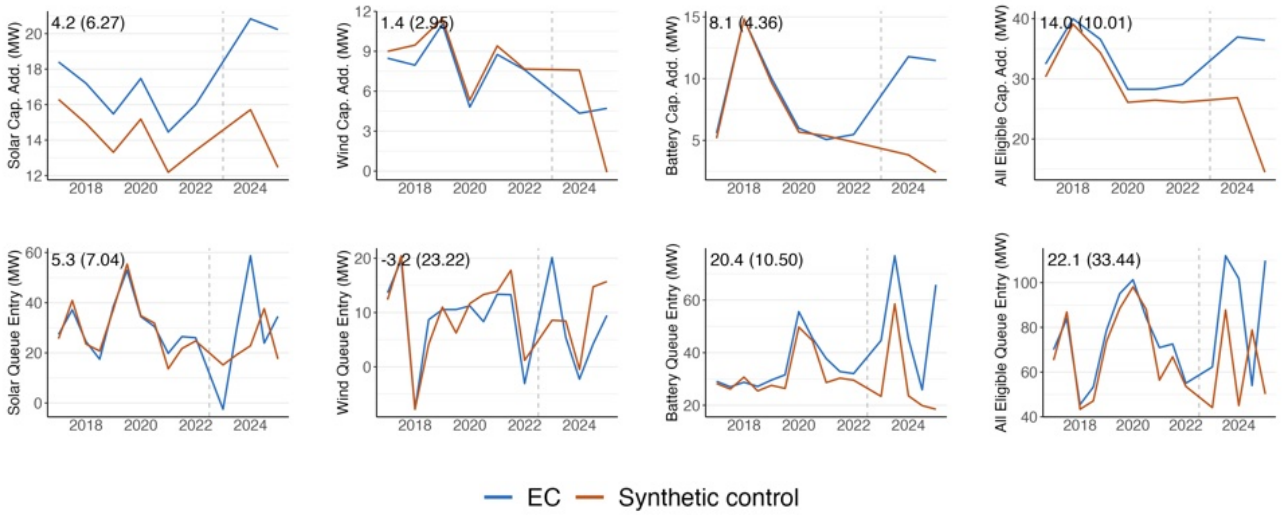



Figure D.2: Capacity addition (top row) and queue entry (bottom row) estimating SDID Equation 3, with adding a state-by-year fixed effect in addition to the grid-operator-by-year fixed effect.

## D.1 State-by-year Fixed Effects and Dropping California

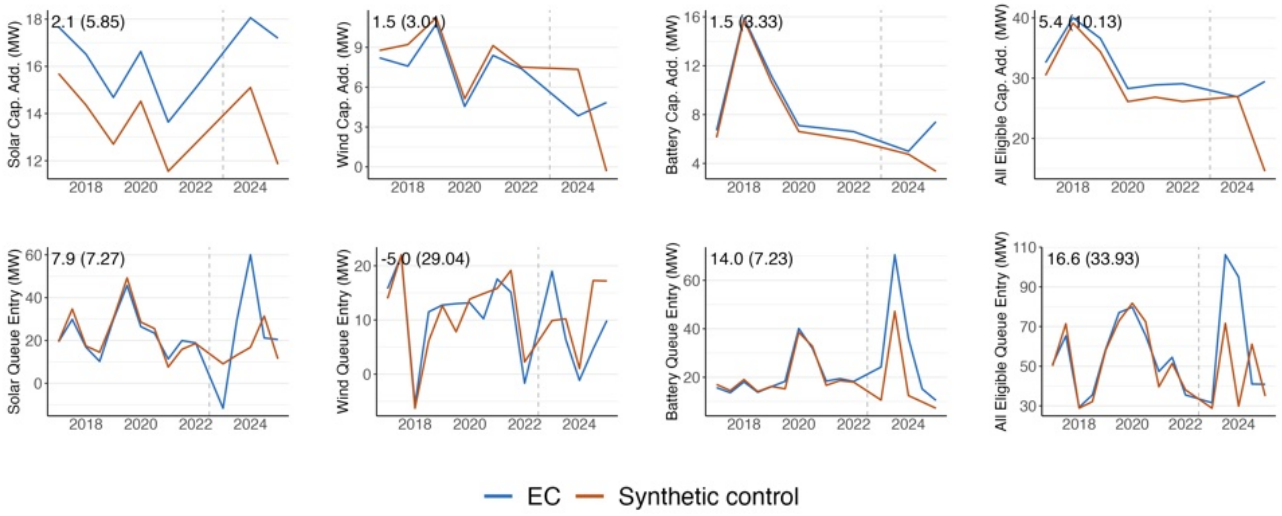



Figure D.3: Capacity addition (top row) and queue entry (bottom row) estimating SDID Equation 3, with adding a state-by-year fixed effect in addition to the grid-operator-by-year fixed effect, and also dropping California.

# E Supplementary Figures for Project-level Analyses

## E.1 Interconnection Costs

I estimate the effect of the subsidy on accepted interconnection costs by estimating the equation,

$$ICCost_{ist} = \tau_c D_{ist} + \gamma EC_i + \alpha_{gt} + \varepsilon_{ist},$$

where $D_{ist}$ is one if study $s$ of project $i$ is in an Energy Community in 2023 or later and zero otherwise, and $EC_i$ is one if project $i$ is in an Energy Community. There is no unit fixed effect because the data are largely cross-sectional; only a few projects have studies before and after the policy. $ICCost_{ist}$ is the interconnection cost. Like the box-plots, the sample includes only cost offers after which projects do not withdraw. $\tau_c$ is the estimated subsidy capitalization into interconnection cost WTA.

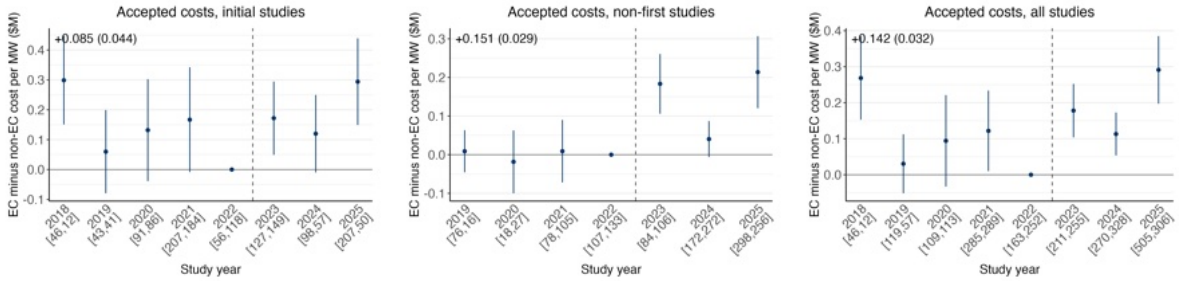
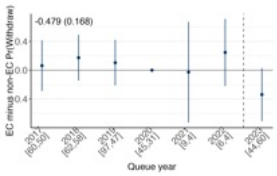
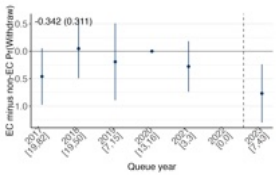



Figure E.1: Event studies for data underlying Figure 7, separating initial studies and non-first studies. Event studies include study year and technology fixed effects. Capacity-weighted. Numbers in brackets are number of project observations in Energy Communities and non-Energy Communities, respectively. Standard errors clustered at the county level.

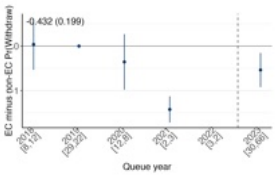
## E.2 Probability of Queue Withdrawal




(a) Solar

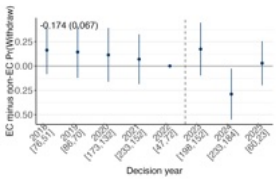



(b) Wind

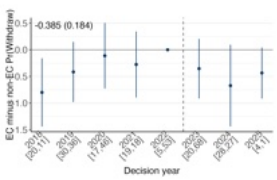



(c) Battery and other

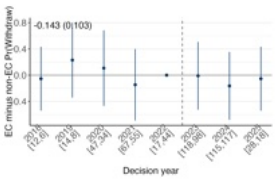
Drop inframarginal, initial studies




(d) Solar

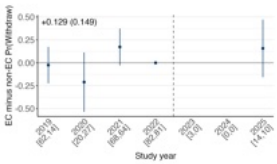



(e) Wind

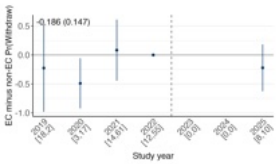



(f) Battery and other

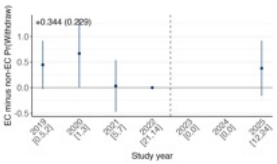
Keep inframarginal, initial studies




(g) Solar

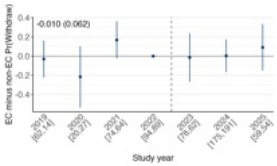



(h) Wind

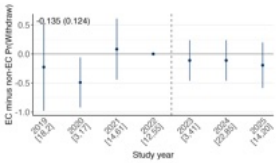



(i) Battery and other

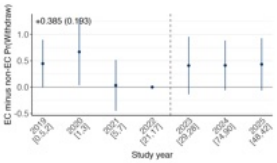
Drop inframarginal, non-first studies




(j) Solar




(k) Wind



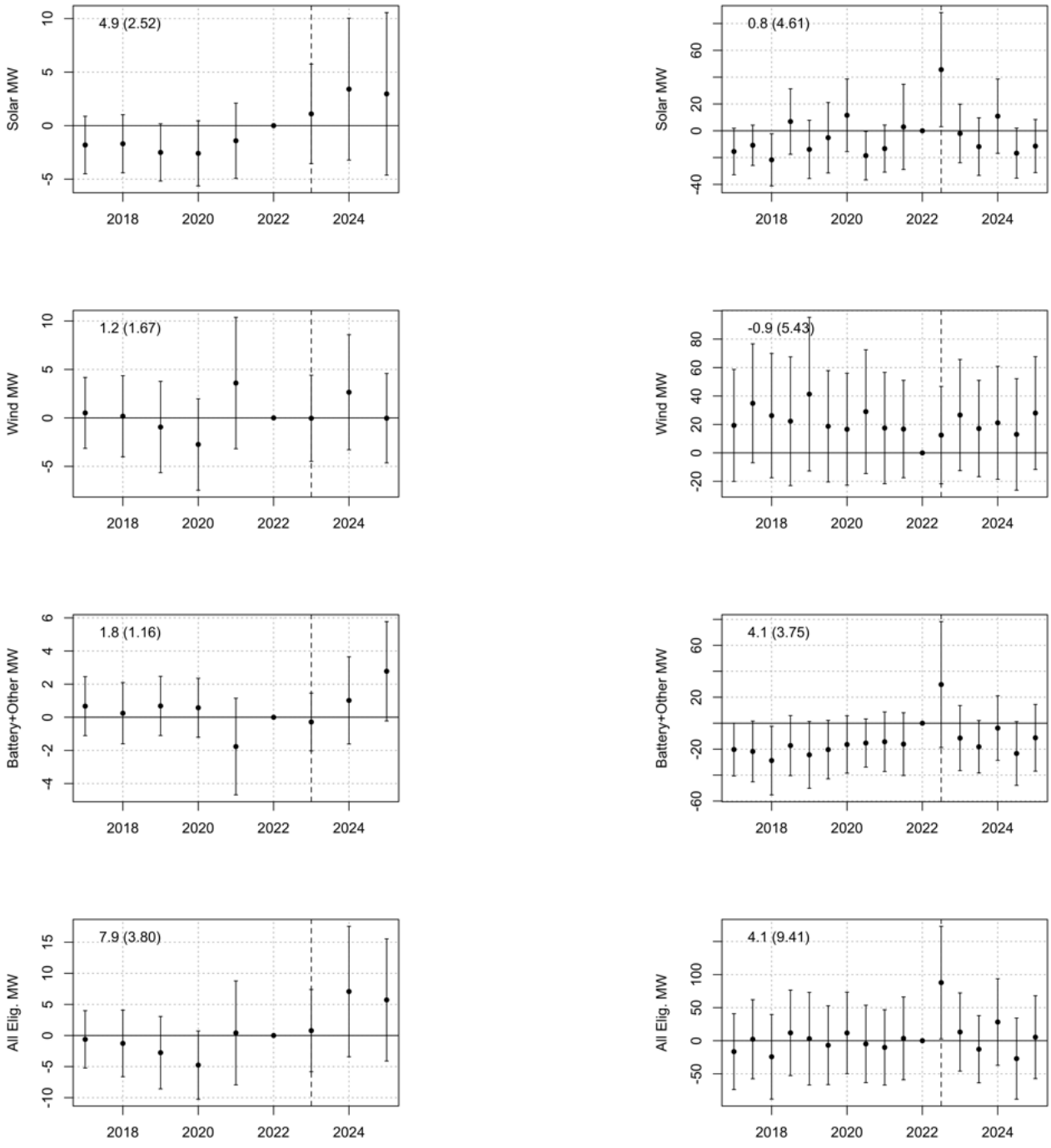

(l) Battery and other

Keep inframarginal, non-first studies

Figure E.2: Figures E.1 and **??**, subsetting the sample on technology type. Pooled coefficient and standard errors reported at the top left. Standard errors are clustered at the county level. Studies for battery projects are rare before 2019, so I fix 2019 to be the reference year.

# F Spillover Effects

## F.1 Estimation with Treatment-Neighbors as the Treated Group




(a) Capacity additions

(b) Queue entry

Figure F.1: Spillover effect event studies and estimates with Energy Community neighbors as the treated group. Pooled treatment effect at the top-left with standard errors clustered at the county level.

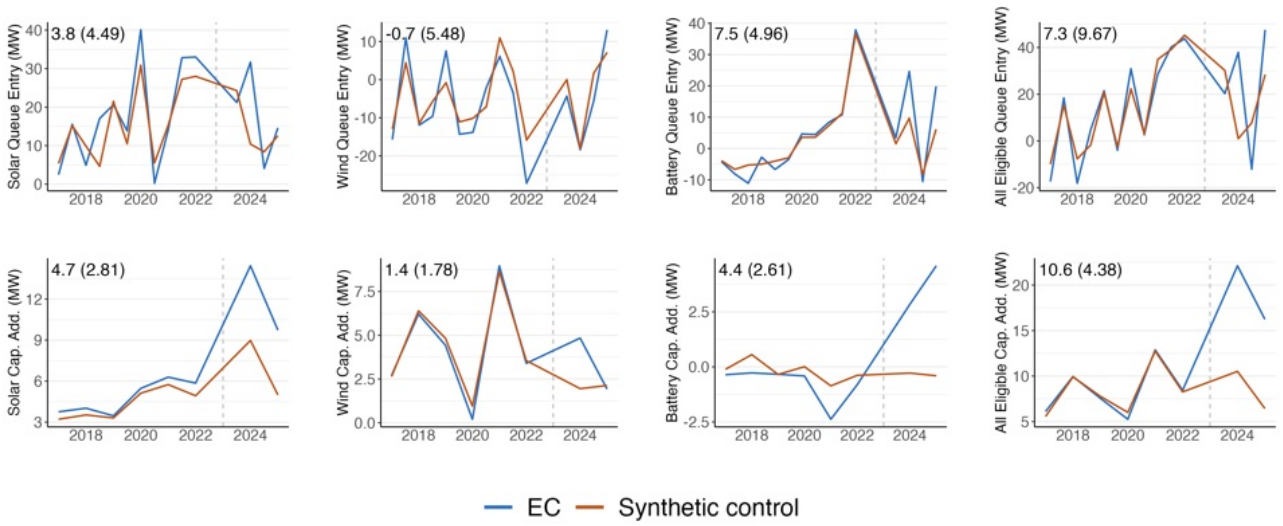



Figure F.2: SDID results for queue entry (top row) and capacity addition (bottom row). Analyses restrict set the "treated" group to counties that neighbor Energy Communities from in the donor pool of controls of the primary specification. Energy Communities are dropped. The donor pool of controls for this specification are counties that are separated from Energy Communities by two or more other counties (i.e., third ring of counties around the Energy Communities and farther).

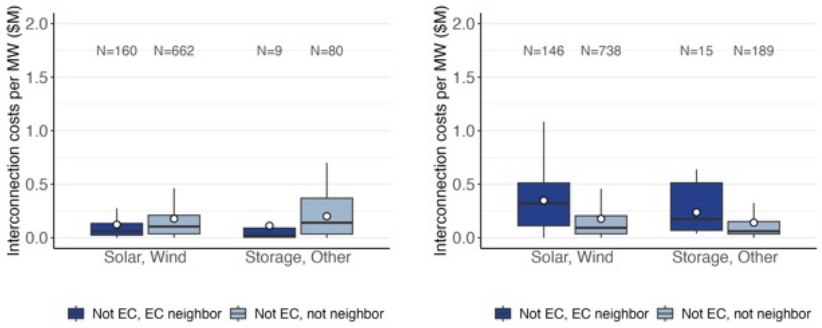



Figure F.3: Interconnection costs per MW by technology before and after the Energy Community Bonus was enacted in 2023. The dark blue are neighbors of Energy Communities that are non-Energy Communities, meeting all of the restrictions that qualify a control county. The light blue are non-neighbor counties.

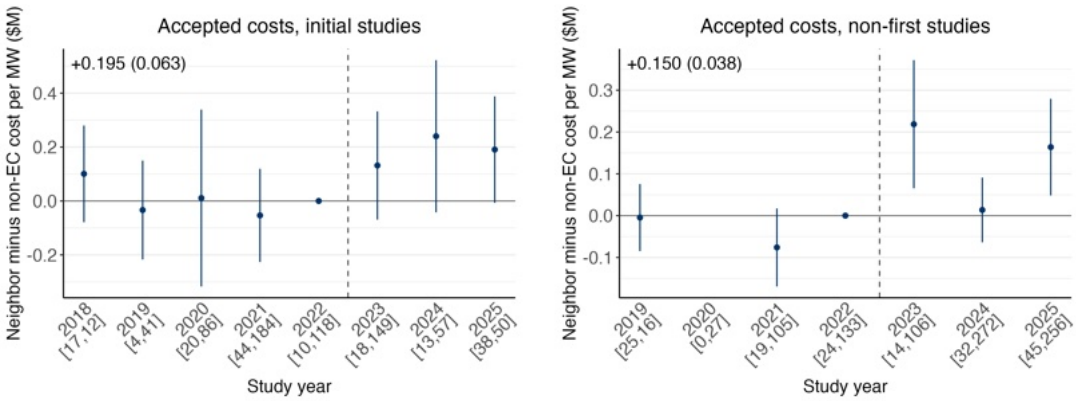
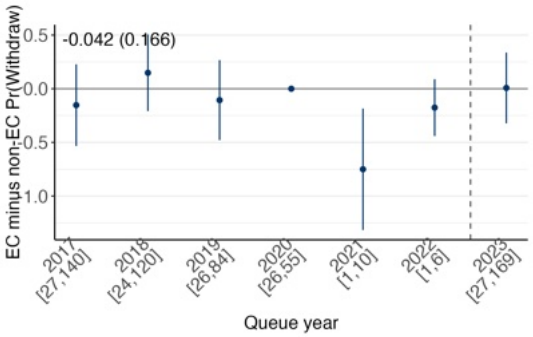
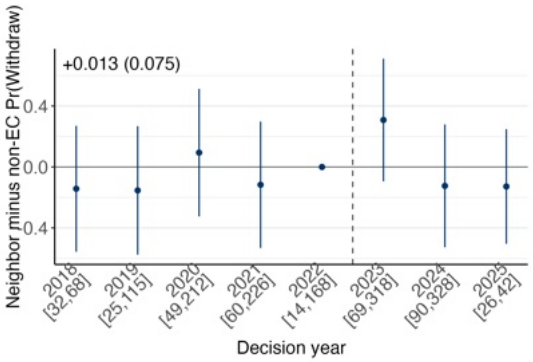



Figure F.4: Event studies for data underlying Figure F.3. Event studies include study year and technology fixed effects. Capacity-weighted. Numbers in brackets are number of project observations in Energy Communities and non-Energy Communities, respectively. Standard errors clustered at the county level.

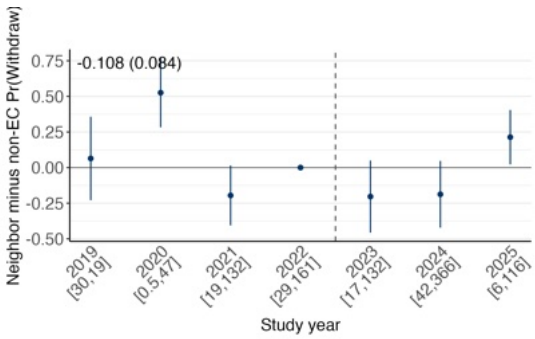



(a) Initial study, drop inframarginal

(b) Non-first studies, drop inframarginal (NA)

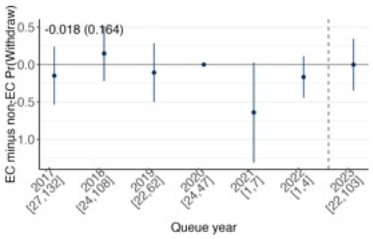


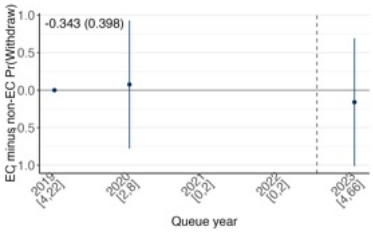

(c) Initial study, keep inframarginal




(d) Non-first studies, keep inframarginal

Figure F.5: Event studies of probability of withdrawal after interconnection study for neighbors of Energy Communities comapard to non-neighbors. NA indiciates insufficient post-period observations to estimate. Capacity-weighted. Numbers in brackets are number of project observations in Energy Communities and non-Energy Communities, respectively.

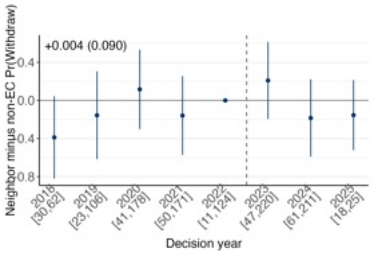


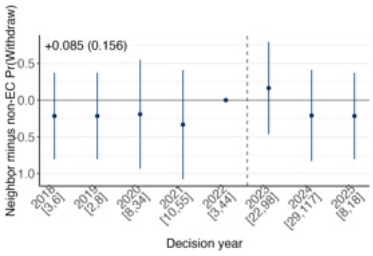




Drop inframarginal, initial studies

(a) Solar, wind

(b) Battery and other




Keep inframarginal, initial studies

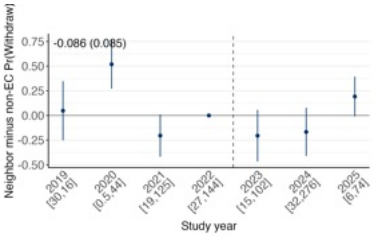
(c) Solar, wind

(d) Battery and other

Drop inframarginal, non-first studies

(e) Solar, wind (NA)

(f) Battery and other (NA)




Keep inframarginal, non-first studies

(g) Solar, wind

(h) Battery and other (NA)

Figure F.6: NA indicates there are 2 or fewer observations in the post-period for the treated neighbors. Capacity-weighted. Numbers in brackets are number of project observations in Energy Communities and non-Energy Communities, respectively.

## F.2 Equilibrium Price Spillover Theoretical Model

Consider a stylized market for electricity in the long run. There is an aggregate demand for electricity that depends on price, $x(p)$.I assume demand is decreasing in price. Since this is a static, long-run model, the relevant price $p$ is some aggregator of current and future electricity prices. Each county $j$ supplies electricity from new renewable capacity additions $y_j^a(p)$, new fossil fuel additions $f_j^a(p)$, and baseline renewable and fossil fuel capacity $y_j^b(p)$ and $f_j^b(p)$. All supply functions are assumed increasing in price.

The market clears when aggregate demand for new capacity equals the sum of new capacity supplied by each county,

$$x(p) = \sum_j y_j^a(p + sEC_j) + y_j^b(p) + f_j^a(p) + f_j^b(p),$$

where $t$ is a subsidy awarded if county $j$ is an Energy Community, captured by the indicator $EC_j$.

Now consider a marginal change to subsidy $s$. Differentiating the market-clearing equation with respect to the subsidy yields the equilibrium change in price,

$$p' = \frac{\sum_j y_j^{a\prime} EC_j}{x' - \left(\sum_j y_j^{a\prime} + y_j^{b\prime} + f_j^{a\prime} + f_j^{b\prime}\right)} < 0,$$

where prime on price indicates its partial derivative with respect to the subsidy, and on supply and demand indicate their partial deriviatives with respect to price. The inequality follows from assuming demand slopes down and supply slopes up. Intuitively, subsidizing supply lowers the equilibrium price.

Denote aggregate supply functions by dropping the $j$ subscript, and denote quantities demanded and supplied as the same variables without parentheses (i.e., $x$ total quantity demanded, $y_j^a$ the total subsidy-eligible new electricity in county $j$). Then, we can write the equilibrium change in price in terms of elasticities and the share of total electricity that comes from new supply in Energy Communities,

$$\begin{aligned} p' &= \frac{\sum_j y_j^{a\prime} EC_j}{x' - y'} \\ &= \frac{\sum_j \frac{y_j^a}{p} \varepsilon_{y_j^a} EC_j}{\varepsilon_x \frac{x}{p} - \frac{y^a}{p}\varepsilon_{y^a} - \frac{f^a}{p}\varepsilon_{f^a} - \frac{y^b}{p}\varepsilon_{y^b} - \frac{y^b}{p}\varepsilon_{f^b}} \\ &= \frac{\sum_j \theta_{y_j^a} \varepsilon_{y_j^a} EC_j}{\varepsilon_x - \theta_{y^a}\varepsilon_{y^a} - \theta_{f^a}\varepsilon_{f^a} - \theta_{y^b}\varepsilon_{y^b} - \theta_{f^b}\varepsilon_{f^b}}. \end{aligned}$$

The first equality applies the definition of aggregate supply. The second equality substitutes the elasticity of demand $\varepsilon_x = x' \frac{p}{x}$, the elasticity of aggregate supply $\varepsilon_y = y' \frac{p}{Y}$, and the county-level subsidy-eligible elasticities of supply $\varepsilon_{y_j^a} = y_j^{a\prime} \frac{p}{y_j^a}$. The last equality comes from the fact that aggregate quantity supplied $Y$ equals aggregate demand $X$ in equilibrium, where $\theta_{y_j} = y_j / X$ is the share of *total* electricity (or total capacity in a capacity market) coming from new renewables in county $j$.

Finally, consider an estimate of subsidy treatment effects that comes from a subsidized county, $j = E$ to an unsubsidized county $j = N$. Since these areas are comparable, I assume the counties have equal supply functions for new electricity. Such a comparison would over-estimate the true subsidy effect by the multiplier

$$\begin{aligned}
\frac{dy_E - dy_N}{dy_E} &= 1 - \frac{dy_N}{dy_E} \\
&= 1 - \frac{g_N^{a\prime} p'}{g_E^{a\prime}(p'+1)} \\
&= 1 - \frac{p'}{(p'+1)} \\
&= 1 - \frac{\sum_j \theta_{y_j^a} \varepsilon_{y_j} EC_j}{\varepsilon_x - \theta_{f^a} \varepsilon_{f^a} - \theta_{y^b} \varepsilon_{y^b} - \theta_{f^b} \varepsilon_{f^b} - \sum_j \theta_{y_j^a} \varepsilon_{y_j^a} (1 - EC_j)}
\end{aligned}$$

The first and second lines are algebra and substituting $dg_N = g_N' p'$ and $dg_E = g_E'(p'+1)$, the third comes from the assumption the two counties are comparable, and the last line substitutes in $p'$ and re-arranges. $\theta_{f^a}$ and $\varepsilon_{f^a}$ and the share of total electricity from and elasticity of new fossil fuel additions. The other variables are defined similarly.

These parameters are challenging to estimate in practice, largely because the price $p$ is some aggregator of present prices and developers' expected future prices.[27] However, reasonable assessment suggests the multiplier is close to one.

Even without knowing elasticities, the multiplier is likely close to one simply because capacity additions in Energy Communities are small relative to the entire US electricity grid. This is captured by the shares in the numerator of Equation **??**, $\theta_{y_j^a}$. My estimates below suggest the Energy Community Bonus causes about 40 GW of new clean electricity capacity added to the grid per year. This is a lot, but even if repeated for many years, it is small relative to the full grid, which had about 1,281 GW of nameplate capacity in 2025. These numbers are nameplate capacities, so the intermittent renewable generators and batteries captured by $\sum \theta_{y_j^a} EC_j$ are an even smaller share than 40/1281.

[27] If one could come up with a measure of price expectations, the Energy Community Bonus would provide sufficient variation as an instrument to estimate both the supply and demand elasticities, using the framework in Zoutman et al. (2018). I leave this for future work, presenting this model for argument's sake.

The multiplier is also close to one if *long-run* electricity demand and supply are elastic ($\varepsilon_x$ is very negative, and $\varepsilon_{y^b}$, $\varepsilon_{f^b}$, $\varepsilon_{y^a}$, $\varepsilon_{f^a}$ are very positive). There is limited evidence of long-run electricity demand elasticity, but Buchsbaum (2022) finds a highly elastic long-run residential electricity demand in California. As for supply elsticities, usually renewable electricity supply is considered inelastic. However, in the long-run there is more room for adjustment, and the limited evidence suggests a more elastic supply. For example, Johnson (2014) finds a long run elasticity of renewable electricity capacity of 2.67. Combined with the relatively small share of generation from Energy Community new capacity additions, these long-run elasticities suggest limited scope for price-driven spillovers.

Note that an alternative way to consider this model is with an inelastic baseline capacity. The key intuition holds. In particular, the numerator still includes $\sum_j \theta_{y_j^a} EC_j$, the share of *total, grid-wide* electricity capacity that is from new additions in Energy Communities, as well as relatively large long-run supply and demand elasticities.

# G Emissions and Welfare Model

## G.1 Models for Intuition on Place-based Clean Electricity Subsidies

Consider an aggregate demand for electricity $x(p)$, measured in $MWh$, which yields societal utility $u(x(p))$, where $x' < 0$, $u' > 0$ and $u'' < 0$. The economy also features a numeraire $x_0$ that delivers linear utility with a price normalized to one.

$J$ heterogeneous sources of electricity supply electricity capacity in perfect competition with supply functions $y_j(p)$, measured in $MW$, with $y_j' > 0$ for all $j$. Some sources are more productive than other areas (i.e., sunnier and windier), such that $y_j$ capacity produces $a_j y_j$ units of electricity, where $a_j$ is a capacity factor. Source-specific cost functions $c_j(y_j(p))$ have $c' > 0$ and $c'' > 0$.

Finally, electricity produces a source-specific externality $\phi_j$. The market clears when $x(p) = \sum_j a_j y_j(p)$.

Baseline welfare is given by

$$W = U(x(p)) - px(p) + \sum_j pa_j y_j(p) - c_j(y_j(p)) + \phi_j a_j y_j(p).$$

With a social planner using a production subsidy, welfare is

$$W^p = U(x(p)) - px(p) + \sum_j (p + s_j^p) a_j y_j(p + s_j) - c_j(y_j(p + s_j^p)) + \phi_j a_j y_j(p + s_j^p) - s_j^p a_j y_j,$$

where $s_j^p$ is a source-specific subsidy (or tax) per MWh. With an investment subsidy, welfare becomes

$$W^i = U(x(p)) - px(p) + \sum_j pa_j y_j(p) - c_j(y_j(p)) + s_j^i y_j + \phi_j a_j y_j(p) - s_j^i y_j,$$

where $s_j^i$ is a source-specific subsidy (or tax) per MW. This clearly cancels out in the welfare function, but affects the result as it enters the producers' optimal capacity choice.

Differentiating the welfare function with respect to the subsidy for some source $k$ yields first order conditions to obtain the optimal subsidies,

$$\frac{\partial W^p}{\partial s_k^p} = (\phi_k - s_k^p) a_k \frac{\partial y_k}{\partial s_k^p} + \sum_{j \neq k} (\phi_j - s_j^p) a_j \frac{\partial y_j}{\partial s_k^p} = 0,$$
$$\frac{\partial W^i}{\partial s_k^i} = (\phi_k a_k - s_k^i) \frac{\partial y_k}{\partial s_k^i} + \sum_{j \neq k} (\phi_j a_j - s_j^i) \frac{\partial y_j}{\partial s_k^i} = 0.$$

The second-best optimal production subsidy is $s_k^{p*} = \phi_k$ and investment subsidy is $s_k^{i*} = \phi_k a_k$ for all $k$. That is, the optimal production subsidy does not depend on geography, while the optimal investment subsidy does.

In the case that subsidies are for some reason constrained to be sub-optimal, however, the change in welfare caused by a unit-increase in production subsidy still depends on geography $a_k$. In the case of under-subsidizing, $s_k^p < \phi_k$, a marginal subsidy increase to a more productive source (higher $a_k$) yields a higher social benefit. In the case of over-subsidizing, $s_k^p > \phi_k$, and a marginal subsidy increase to a more productive patch is more distortionary and implies greater social cost.

The point is, for investment subsidies, moving from a uniform to a place-based subsidy may improve or worsen welfare because the initial subsidy is obviously sub-optimal under any heterogeneity in externalities or geography. For production subsidies, spatially differentiating the subsidy by geography is not clearly welfare improving. However, in the case that the baseline uniform subsidy is too low, differentiating by geography still has ambiguous welfare effects compared to some other capacity-neutral or fiscally-neutral subsidy.

## G.2 Emissions and Tax Data and Calculations

### G.2.1 Emissions Abated by Clean Electricity

I rely on the National Lab of the Rockies (NLR) Cambium 2024 model to measure the emissions abated by a marginal clean electricity generator (Gagnon et al., 2024). In particular, I use their "Mid-Case" scenario, short-run marginal emissions projections. The short-run marginal emissions are the emissions that would occur from a marginal increase in a region's load. I use Cambium's busbar adjustment factor to convert to the marginal emissions that would arise from a marginal increase in the region's generation. The only differene between these is the assumed transmission losses from generation or distribution. The data are at the "pzone" level, which are clusters of counties. There are 133 pzones. The marginal emissions data are zone-by-hour-of-year projections from 2025 to 2050 at five-year intervals. I linearly interpolate between the five year intervals at the pzone-by-hour-of-year level to create annual data.

I use Cambium's short-run marginal emissions instead of their long-run marginal emissions. Their long-run marginal emissions product models a country-wide shock to electricity load and reports its effect on local emissions. This is a fundamentally different thought-experiment from that of the short-run marginal emisssions rate, and the short-run marginal emission is what matches the nationwide emissions effect due to a marginal generation increase.

I choose to use a modeled marginal emission, instead of the many empirical estimates available

the literature (e.g., Siler-Evans et al. (2012), Graff-Zivin et al. (2014), Holland et al. (2022), and Holland et al. (2024)). While there is a strong justification to use the empirical estimates for present-day empirically estimated marginal emissions, the marginal emission is likely to change in the future. I assume new capacity built in 2025 is online for 26 years until 2050. Although marginal emissions are stable relative to average emissions, the forecast is better suited than the empirical estimates for such a long time horizon (Holland et al., 2022).

My outcome variables are measured in MW of nameplate capacity, but it is MWh of electricity generation that abates electricity. I convert MW to MWh using county-level capacity factors for solar and wind technologies. This depends on location and the type of wind turbine or solar panel used. The 2024 County-Level Hourly Renewable Capacity Factor Dataset, which underlies NREL's ReEDS model (Sergi et al., 2024). The wind data county-by-hour capacity factors for 15 weather years, 2007 to 2013 and 2016 to 2023. There are values for multiple different "resource classes" in each county. I assume that the wind generation takes the capacity factor of the best resource class in the county. I average this value across the 15 weather years for the final measure of each county's hour-of-year wind capacity factor. I use a similar procedure to obtain solar capacity factors.

For battery capacity factors, I assume all batteries added to the grid charge and discharge 4 MWh per MW per day. Cambium provides projections for wholesale electricity prices. I interpolate to obtain annual prices in the same way as I do marginal emissions. I assume that batteries charge when it is least expensive, emitting the marginal emission, and discharge when electricity is most expensive, abating the marginal emission. I use random tie-breakers in the relatively rare instance that there are many hours in the day with the same electricity price.

Finally, I convert all the emissions abated and emitted to present discounted value (PDV) by county-technology, again assuming the capacity survives for 26 years beginning in 2025 and ending in 2050. I assume a 2 percent social discount rate on all future emissions. For emissions outcomes, this is the final metric.

To value the emissions, I use the social cost of carbon (SCC) from 2 percent discount rate scenario of EPA (2023). This value increases into the future, and I value the emissions abated at future dates at its future SCC level before convertintg the monetized emissions to a PDV. The outcomes are shown in Figure **??**.

### G.2.2 Tax Credit Values and Policy Changes

For every technology, I calculate the value of the PTC per MW of capacity and the value of the ITC per MW of capacity in every county. I assume the maximum of those two values is the

value of the tax credit per MW by technology and locations.

The PTC per MW requires calculating how much electricity a MW of each technology in each location would generate for its first 10 years online. I use the same capacity factors from the previous section to obtain this stream of future electricity generation. I assume all projects meet the wage and apprenticeship requirements, such that the value of the PTC in 2023 is 2.75 cents per kWh. The tax credit is inflation-adjusted. I increase the tax credit at 2.5 percent per year, based on the 30-year expected inflation in 2025 (FRED, 2025). Lastly, I must discount the tax credit to a PDV such that its comparable to the ITC, which accues immediately. I assume developers use a private discount rate of 8 percent to value the subsidy.

The ITC calcualtion is more straightforward. I obtain the cost per MW to build a new wind, solar, or battery generator from National Lab of the Rockies' Annual Technology Baseline (NLR, 2024). I use the capital expenditure value, which includes "overnight costs" as well as grid construction costs. The capital expenditures depend on the type of generator built (e.g., wind turbine size or solar panel capacity factor), which on a location's resource class. ATB models the profitability of each possible decision, and assume projects choose the most profitable generator type for each resource class in each location. Like the capacity factor calculation, I assume generators take the best resource class. The ITC is a percent of this capital expenditure and does not need to be discounted or inflated to compare to the PTC.

Lastly, to calculate technology shares for each county, I use queue entry data from 2023 to 2025. For counties with queue entries, I set the technology shares of solar, wind, and batteries to the observed shares of each of those three technologies in the queue. For counties without queue entries, I assign the state share of each technology. For states with no queue data, I use the national share of each technology.




(a) Wind (b) Solar (c) Battery

Figure G.1: Tax credit choices for the baseline ITC and PTC (top row), with the Bonus (middle row), and the Energy Community Bonus value (bottom row). NA counties are those considered unsuitable for renewable electricity development in the capacity factor data, for example for being protected lands.

## G.3 Formal Derivation of the Welfare Expression in Equation 5

This section presents the static model that derives Equation 5. The set up builds on Jacobsen et al. (2020), who base thier model on Harberger (1964).

Consider a representative consumer of county-by-technology electricity capacity $x_j$, measured in MW (e.g., MW of solar capacity in New Haven county), and a numeraire $x_0$ with price set to 1. The consumer maximizes its utility

$$\max_{x_0,x_1,...,x_J} U(x_1,...,x_J) + x_0 \qquad \text{s.t. } x_0 + \sum_j p_j x_j \leq M,$$

where there are $J$ total county-technology combinations, and $M$ is an exogenously specified income, and $p_j$ is the price of $x_j$. Utility is assumed once differentiable, weakly increasing in all $x_j$.

A representative producer operates in a perfectly competitive environment to maximize profit

$$\max_{y_1,...,y_J} \sum_j (p_j + t_j) y_j - C(y_1,...,y_J),$$

where $y_j$ is the supplied MW of county-technology $j$, $C_j$ a total cost function, and $t_j$ is a subsidy for $j$. The market clears when $x_j = y_j$ for all $j$.

Welfare is the sum of consumer surplus and producer surplus, minus the transfer required for the subsidies, plus an externality from consuming a unit of $x_j$, denoted $\phi_j$:

$$\begin{aligned} W &= U(x_1,...,x_J) + M - \sum_j p_j x_j + \sum_j (p_j + t_j) x_j - C(x_1,...,x_J) + \sum_j \phi_j x_j - \sum_j t_j x_j \\ &= U(x_1,...,x_J) + M - C(x_1,...,x_J) + \sum_j \phi_j x_j. \end{aligned}$$

The first line subsitutes the consumer's budget constraint for $x_0$ and $x_j$ for $y_j$ according to the market clearing equation. The second line simplifies.

Toward proving Equation 5, first note that the change in welfare for a marginal change in the subsidy for a particular county-technology is

$$\frac{\partial W}{\partial t_j} = \sum_{k=1}^{J} \left( \frac{\partial U_k}{\partial x_k} - \frac{\partial C_k}{\partial x_k} + \phi_k \right) \frac{\partial x_k}{\partial t_j} = \sum_{k=1}^{J} \left( p_k - (p_k + t_k) + \phi_k \right) \frac{\partial x_k}{\partial t_j} = \sum_{k=1}^{J} \left( \phi_k - t_k \right) \frac{\partial x_k}{\partial t_j}.$$

The second equality follows from the consumer and producer's first order conditions in perfect competition.

Now, consider a change from some arbitrary tax vector $\tau = (\tau_1, ..., \tau_J)$ to another arbitrary tax vector $\tau'$. Let $\tilde{t}_j = (1-\rho)\tau_j + \rho\tau'_j$ for all $j$. Differentiating the welfare function with respect to $\rho$ and integrating the result from $\rho = 0$ to $\rho = 1$ yields Equation 5.

$$\begin{aligned}
\frac{\partial W}{\partial \rho} &= \sum_j \sum_k \left( \frac{\partial U_j}{\partial x_j} - \frac{\partial C_j}{\partial x_j} + \phi_j \right) \frac{\partial x_j}{\partial t_k} \frac{\partial t_k}{\partial \rho} \\
&= \sum_j \sum_k \left( \phi_j - \tilde{t}_j \right) \frac{\partial x_j}{\partial t_k} \frac{\partial t_k}{\partial \rho} \\
&= \sum_j \sum_k \left( (\rho - 1)\tau_j - \rho\tau'_j + \phi_j \right) \frac{\partial x_j}{\partial t_k} (\tau'_k - \tau_k).
\end{aligned}$$

The second line follows from the first order conditions as before. The third substitutes in the definition and derivative of $\tilde{t}$. Finally, it is straightforward to integrate and rearrange this expression,

$$\int_0^1 \sum_j \sum_k \left( (\rho - 1)\tau_j - \rho\tau'_j + \phi_j \right) \frac{\partial x_j}{\partial t_k} (\tau'_k - \tau_k) d\rho \quad = \frac{1}{2} \sum_j \sum_k (\phi_j - \tau_j + \phi_j - \tau'_j) \frac{\partial x_j}{\partial t_k} (\tau'_k - \tau_k).$$

# H Aggregate Trends in Capacity Additions and Queue Entry

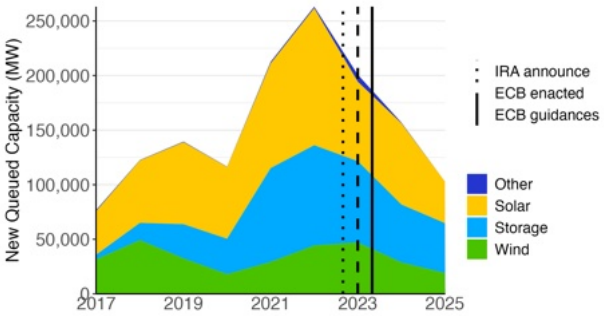
Figure H.1 shows trends in queue entry in the left column and capacity additions in the right column. The top row shows these statistics by technology type, and the bottom by Energy Community status. The graphs only show projects in areas that are Energy Communities and non-Energy Communities, dropping projects with uncertain Energy Community designation.[28]

Much fewer projects complete the queue than begin the queue; the y-axes of the queue entry and capacity addition charts differ by an order of magnitude. Queue entries grow quickly, reach a peak in 2022, and then contract after 2023, likely due to efforts by grid operators (such as the 2022 PJM two-year pause on queue entries) and the FERC 2023 mandate to reform the queue entry processes, both attempting to address the increasing backlog of queue entries. A growing share of capacity in queues is energy storage. This category includes both standalone and hybrid storage projects.

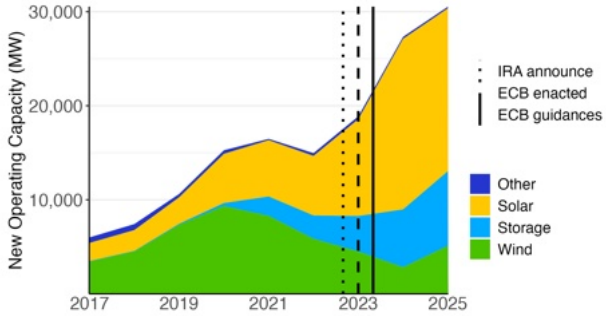
The dotted, dashed, and solid black lines show when the Energy Community Bonus was announced (July 2022), enacted (January 2023), and the guidance that detailed the locations of Energy Communities (April 2023). Energy Communities show a smaller reduction in queue entry after 2023 than do non-Energy Communities. There are more capacity additions in Energy Communities than in non-Energy Communities after 2023, but there is also evidence of pre-trends, motivating a careful empirical analyis.

[28]The Energy Community category includes projects in counties that are (1) completely covered by eligible coal tracts, (2) qualify as a fossil fuel employment Energy Community in all years 2023 to 2025, or (3) contain more than 10 brownfields. The non-Energy Community category are projects in counties with no coal tracts, never meeting the fossil employment definition, and with fewer than 10 brownfields.

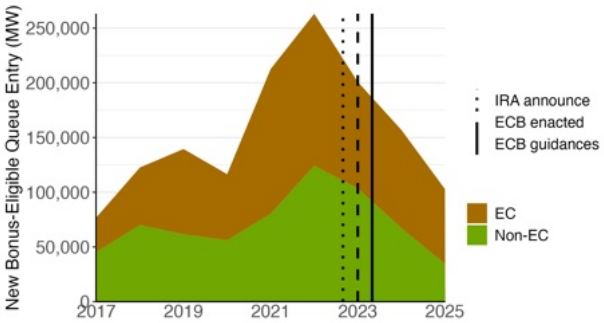
(a) Queue entry by technology.




(b) Capacity additions by technology.




(c) Queue entry by EC status.




(d) Queue completion by EC status.

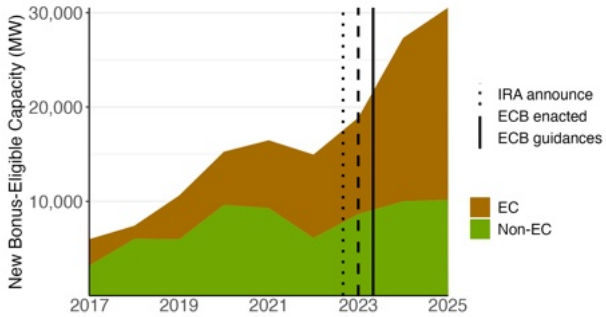



Figure H.1: Panels (a) and (b) chart MW of queue entry and capacity additions over time, respectively, separated by technology. Panels (c) and (d) chart the same MW, separated by Energy Community (EC) Bonus eligibility. All graphs only count MW of technologies eligible for the Inflation Reduction Act subsidies. The graphs also only include projects in the counties that are completely covered by Energy Communities or with no Energy Community designation.